

Trainings ▾

General Information

with Mechanically Actuated Injector

All K38 and K50 engines are turbocharged. All turbochargers on K38 and K50 engines are oil cooled and lubricated. The engine can have two or four turbochargers.

If the engine has four turbochargers, the two that are mounted on the exhaust manifold are the high-pressure turbochargers, and the two that are mounted on the support brackets are the low-pressure turbochargers.

The correct turbocharger assembly **must** be installed. Many turbochargers appear the same physically, but contain different internal parts. If the wrong turbocharger assembly is installed, the performance of the engine will be less than specification, or the engine will be damaged due to excessive pressure and temperature in the cylinders.

K38 and K50 engines can contain turbochargers that are manufactured by Holset®, Brown Boveri®, or AiResearch®. In some cases turbochargers from two different vendors are on the same engine. One of the dataplates that is attached to the turbocharger identifies the vendor, the turbocharger model, and the Cummins® part number. Cummins Inc. identifies the turbocharger by the part number, **not** the model nomenclature.

This manual does **not** contain information to rebuild the turbocharger. See the following turbocharger rebuild manuals:

Manufacturer	Bulletin	Model Number and Description
Holset®	3810243	HC5A Turbocharger Shop Manual
Holset®	3810386	HT100 Series Turbocharger Shop Manual
Brown Boveri®	3810235	Brown Boveri® Turbocharger Manual
AiResearch®	3379091	T18A, Turbocharger Rebuild Manual

Two oil supply hoses are installed on K38 and K50 engines. The length of the hose for KT38, KTA38, and KTA50 engines is 1295 mm [51 in]. The length of the hose for KTTA38 and KTTA50 engines is 1092 mm [43 in].

The turbocharger oil drain assembly for the K38 is shorter on engines with dry type exhaust manifolds than on engines with wet type exhaust manifolds. The turbocharger oil drain assembly for the K50 is the same as that installed on the K38 with a wet type exhaust manifold.

The method for connecting the turbocharger oil drain to the cylinder block was changed. Engines with an engine serial number (ESN) of 33106693 and greater have a flange and an o-ring seal type connection that is secured with a capscrew. Engines with an ESN of 33106692 and lower have a 90 degree fitting with pipe threads.

A revised turbocharger compressor wheel was released for KTA50-G engines in September, 1985, ESN First 33110070.

Turbochargers with a hose nipple on the compressor discharge and a hose type air crossover were released for use in production on the KTA38 engines in September 1986, ESN First 33111500.

A chamfer was added to the diffuser plate on HC5A turbochargers to allow oil drain back to the sump in February, 1987, ESN First 33111870.

A revised profile, turbocharger mounting gasket was released in August, 1987, ESN First 33112713.

A turbocharger oil drain hose that is manufactured of flexible hose is available for engines built previously with a one or two piece steel tube for the oil drain hose. The two piece steel tube that is connected by a hump hose is available for service on engines that have AiResearch® turbochargers **only**.

The KTTA50 low-pressure turbocharger oil drain hose and drain location was changed in 1988, to provide a better downhill return to the cylinder block. The revised oil drain hose routing also requires a revised turbocharger support, which is modified to provide clearance for the new oil drain hose and a different clamp to retain the oil drain hose.

The revised hose drains the oil to drillings on the top deck at the front of the right bank and the left of the rear bank of the cylinder block. The holes used are partially covered by the turbocharger support. This support requires modification or replacement.

The mounting capscrew for the flange and o-ring seal type turbocharger drain hose connection was changed from 0.250 inch -20 UNC to 0.375 inch -16 UNC. The cylinder block with the smaller holes requires a spacer, Part Number 3177195, between the connection flange and the cylinder block at the left bank rear location. If the spacer is **not** installed, an oil leak will result because the o-ring seal will **not** be in the bore correctly.

The flange on the turbocharger oil drain hose that fits to the turbocharger is different for Holset® and AiResearch® turbochargers. The oil drain hose flange is different for the Brown Boveri® type turbocharger. If the model of turbocharger is changed, the oil drain hose **must** be changed.

A K50 turbine housing with a thickened flange between the capscrew holes to reduce flange distortion was released in February 1988, ESN First 33114833. This is for the K50 pulse exhaust manifolds **only**.

Longer studs, nuts, and spacers were released for the K50 pulse exhaust manifold in May, 1989, ESN First 33115948.

The K50 turbocharger mounting flange was completely thickened to improve gasket reliability. The spacer is no longer required for K50 pulse exhaust manifold engines produced in February, 1990, ESN First 33117281.

The K50 pulse exhaust manifold received a boss to allow use of capscrews in place of studs on engines produced from February, 1990, ESN First 33117392.

A steel oil supply fitting replaced the brass fitting to prevent torque loss/loosening on engines produced from February, 1990, ESN First 33117392.

A steel oil supply fitting replaced the brass fitting to prevent torque loss/loosening on engines produced from September 1990, ESN First 33118322.

The Holset® HT100 low-pressure turbocharger replaced the Brown Boveri® RR151 turbocharger for C1800B and K2000 engines in January 1991, ESN First 33119163.

Note : The HT100 turbocharger can not be used to replace the Brown Boveri® Model RR153 that is used on the KTTA50-C1800A engines.

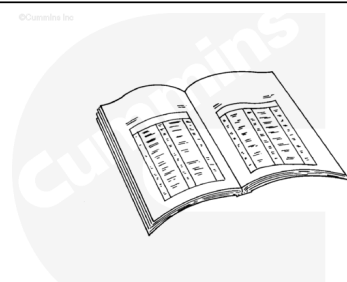
The Holset® HX80 water-cooled turbocharger design was revised in 1997, to allow for the use of GAS fittings on all coolant passage openings in the casting. O-ring seal fittings are less prone to leakage, and are easier to remove and install. The diffuser plate was also changed to a “sunk wheel” design.

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. Refer to the OEM service manual.
- Turn the air OFF to the air starter, if equipped, and tag the valve closed. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Drain the cooling system to a point below the turbochargers. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Remove the coolant supply and return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-041.html>)

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

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To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

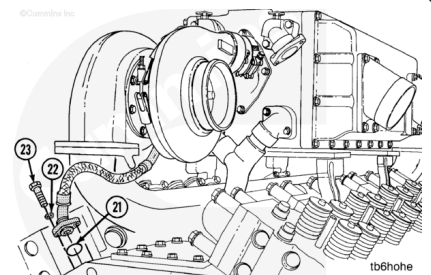
- Disconnect the batteries. Refer to the OEM service manual.
- Turn the air OFF to the air starter, if equipped, and tag the valve closed. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Drain the cooling system to a point below the turbochargers. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Remove the coolant supply and return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-041-tr.html>)
- Remove the oil supply lines. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-046-tr.html>)
- Remove the oil drain lines. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-045-tr.html>)

Remove

with Mechanically Actuated Injector

Note : The turbochargers, mounting hardware, and connections are not the same for all models of K38 and K50 engines. Because the parts are different, procedures are given for removal of the following turbochargers.

- KT38, KTA38, and KTA50 with Center Mount Aftercooler
- KTTA38 Low-Pressure
- KTTA50 Low-Pressure



- K2000E Low-Pressure
- KTTA38, KTTA50, K1500E, K1800E and K2000E High-Pressure, and KT38, KTA38, and KTA50 with Outboard Aftercoolers

Note : The oil inlet and drain hose routing and clamps can be different than shown, depending on the age of the engine.

KT38, KTA38, and KTA50 with Center Mount Aftercooler

Note : The illustrations show a KTA38. The KTA50 has a longer connection between the turbocharger and the aftercooler.

Older engines can have a spacer for the capscrew (23). Keep the spacer for future assembly.

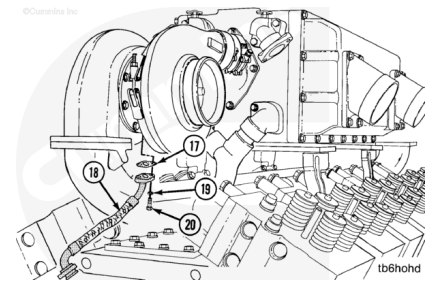
Detach the oil drain hose from the cylinder block. Remove the o-ring seal (21).

Discard the seal.

Detach the oil drain hose from the turbocharger. Remove the gasket (17).

Discard the gasket.

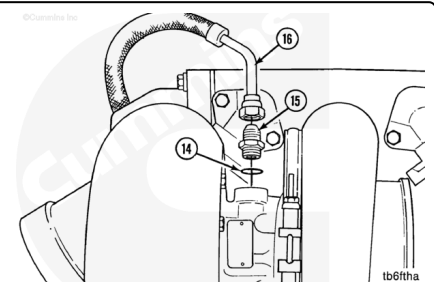
Check the hose (18) for damage. Discard the part if it is damaged.



Detach the oil inlet hose (16). Removal of the fitting (15) and the o-ring seal (14) is **not** required unless they are damaged.

Remove any support clamps from the hose.

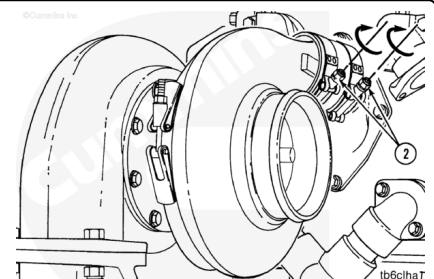
Remove the hose. Check the hose for damage. Discard the part if it is damaged.



Newer engines have a hose and clamps between the turbochargers and air transfer housings.

The clamps are special high strength parts.

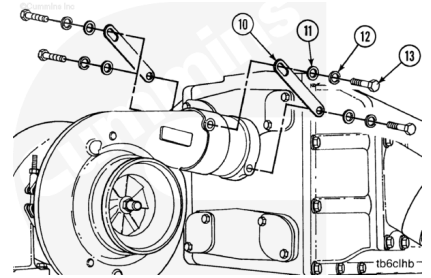
Loosen the hose clamps (2).



Older engines have retaining straps.

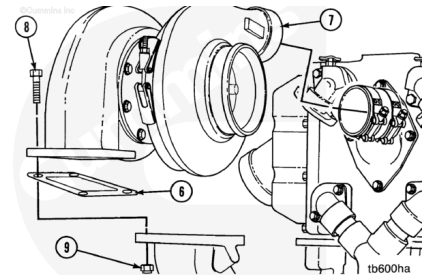
Remove the capscrews and the retaining straps (10).

Check the straps for damage. Discard the part if it is damaged.



⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four capscrews (8) and nuts (9).

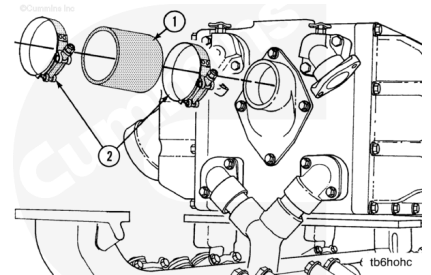
Remove the turbocharger and gasket.

Discard the gasket.

On newer engines, remove the clamps and the hose.

Discard the hose.

Check the clamps for damage. Discard the part if it is damaged.

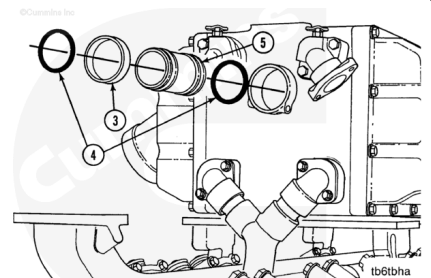


On older engines, remove the tube (5). Remove the o-ring seals (4).

Discard the seals.

Remove the dust seal (3).

If the dust seal is hard or soaked with oil, discard the part.

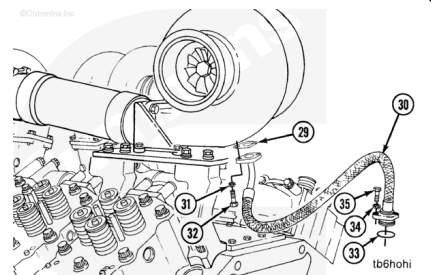


KTТА38 Low Pressure:

Remove the oil drain hose (30). Remove the gasket (29) and the o-ring seal (33).

Discard the gasket and seal.

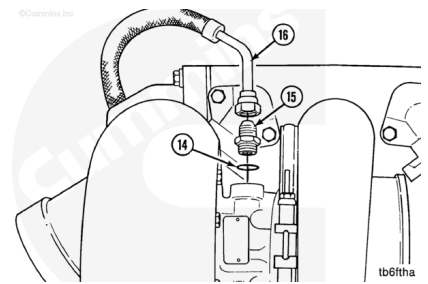
Check the hose for damage. Discard the part if it is damaged.



Detach the oil inlet hose (16). The removal of the fitting (15) and the o-ring seal (14) is **not** required unless damaged.

Remove any support clamps from the hose. Remove the hose.

Check the hose for damage. Discard the part if it is damaged.



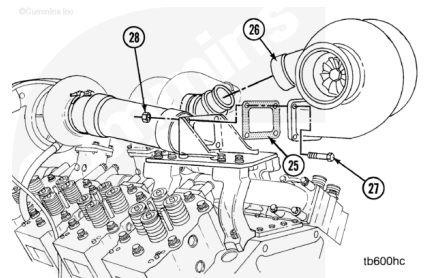
⚠ WARNING ⚠

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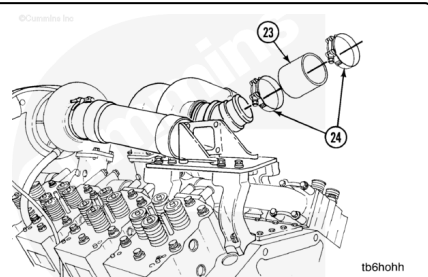
Remove the four capscrews (27) and the nuts (28).

Remove the turbocharger and the gasket.

Discard the gasket.



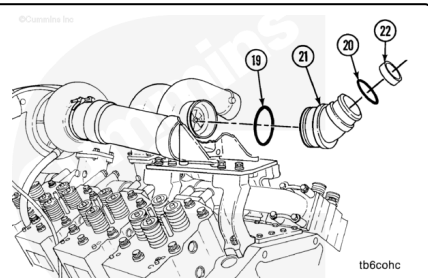
Loosen the hose clamps (24). Remove the clamps and the hose (23) from the high-pressure turbocharger air connection.



Remove the air connection (21) from the high-pressure turbocharger. Remove the o-ring seals (19).

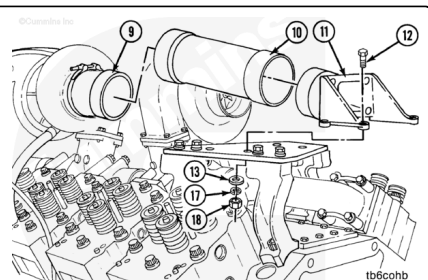
Discard the seals.

Remove the dust seal (22). If the dust seal is hard or soaked in oil, discard the part.

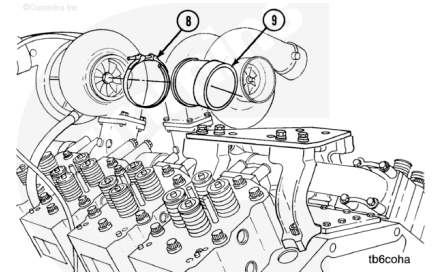


Remove the four capscrews (12) and nuts (18). The heavy washers (13) and capscrews are special heat resistant parts.

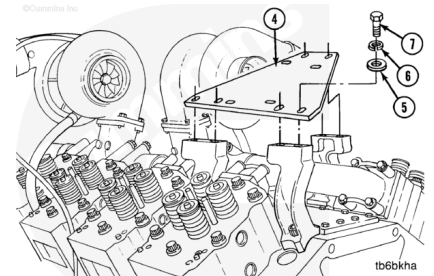
Remove the exhaust connection (11) and the exhaust outlet pipe (10).



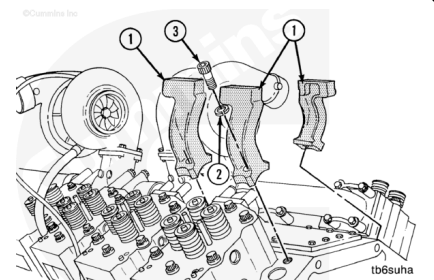
Remove the v-band clamp (8) and the exhaust outlet connection (9).



Remove the six capscrews (7). The heavy flat washers (5) are special. Remove the support plate (4).



Remove the socket head capscrew (3). Remove the turbocharger support legs (1).



KTTA50 Low-Pressure

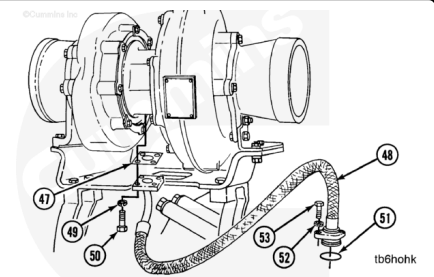
The Brown Boveri®, Model RR151, turbochargers contain metric fasteners for the oil connection plates.

Remove the oil drain hose (48).

Remove the gasket (47) and o-ring seal (51).

Discard the gasket and seal.

Check the hose for damage. Discard the part if it is damaged.

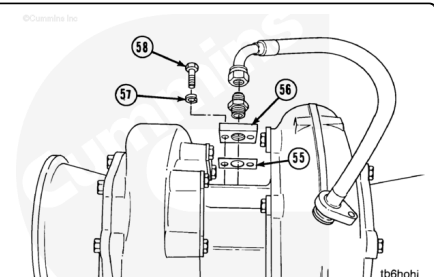


Some Brown Boveri® turbochargers do **not** contain the adapter plates for the oil inlet.

Detach the oil inlet hose. The adapter plates (44) and fittings do **not** require removal.

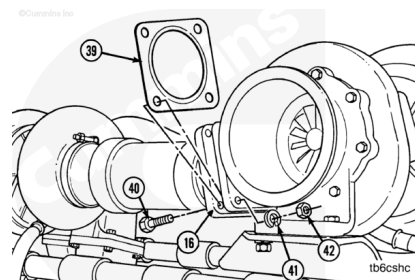
Remove any support clamps. Remove the hose.

Check the hose for damage. Discard the part if it is damaged.

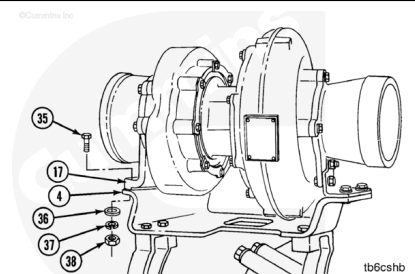


Remove the four capscrews (40) and nuts (42). Remove the gasket.

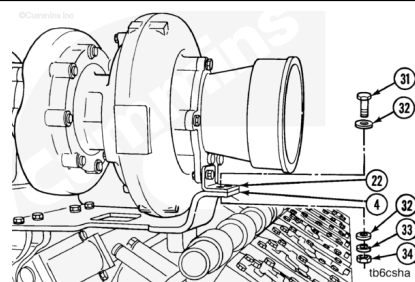
Discard the gasket.



Remove the two capscrews (35) that hold the exhaust side of the turbocharger to the support bracket.

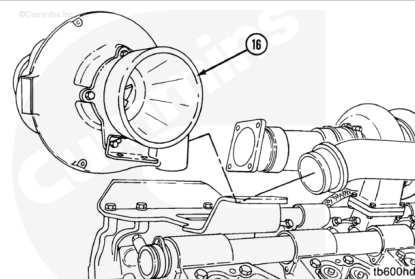


Remove the two capscrews (31) and nuts (34) that hold the intake support bracket to the turbocharger support bracket.



⚠ WARNING ⚠

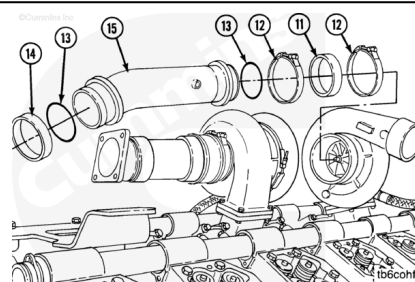
The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



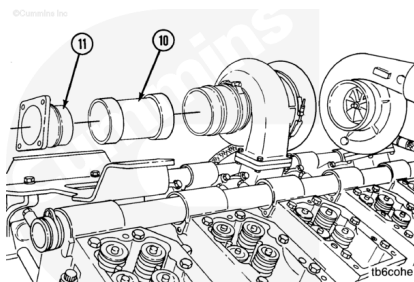
Remove the low-pressure turbocharger (16).

Loosen the hose clamps (12). Pull the intake connection pipe (15) from the high-pressure turbocharger. Remove and discard the o-ring seals (13) and the hose (11). Remove the dust seal.

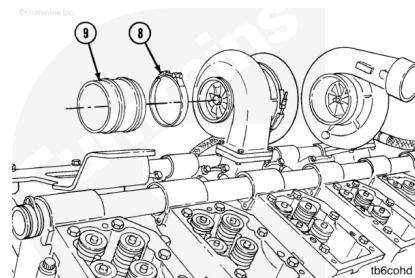
If the dust seal is hard or soaked with oil, discard the part.



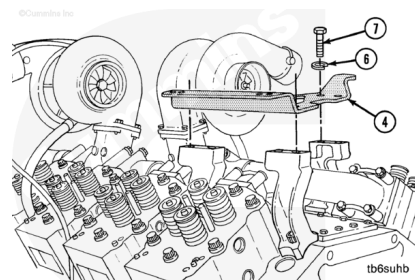
Remove the exhaust connection (11) and outlet pipe (10).



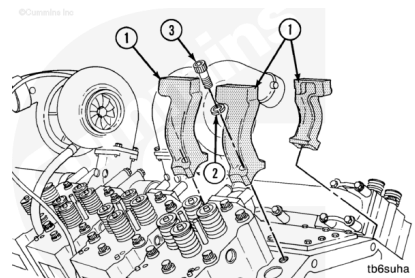
Remove the v-band clamp (8) and the exhaust outlet connection (9).



Remove the six capscrews (7). Remove the turbocharger support bracket (4).



Remove the socket head capscrews (3). Remove the turbocharger bracket support legs (1).

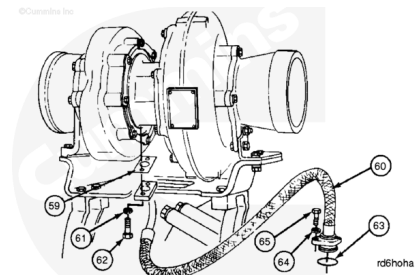


K2000E Low-Pressure:

Remove the oil drain hose (60). Remove the gasket (59) and o-ring seal (63).

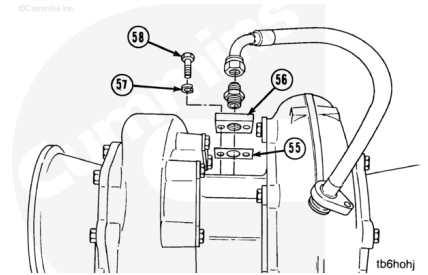
Discard the gasket and seal.

Check the hose for damage. Discard the part if it is damaged.

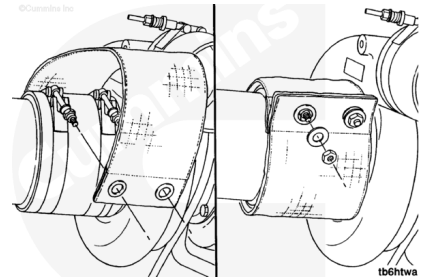


Detach the oil inlet hose. Removal of the fitting, plate, (56) and gasket (55) is **not** required.

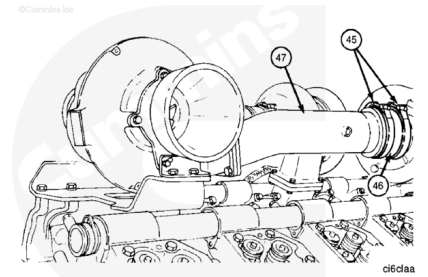
Remove any support clamps from the hose. Remove the hose.
Check the hose for damage. Discard the part if it is damaged.



Remove the heat shield (blanket) from the interstage air hose.



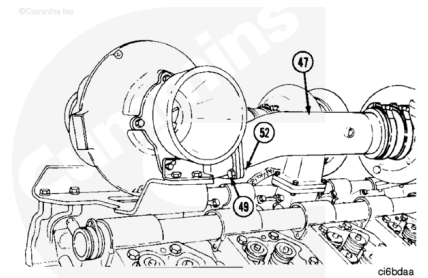
Loosen the two T-bolt clamps (45) on the air intake pipe (47) and the compressor inlet.



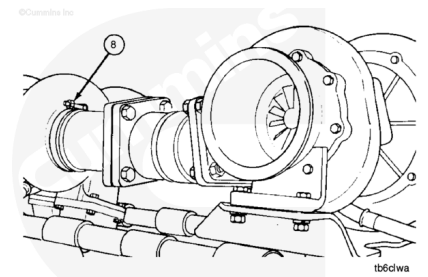
Remove the four capscrews (49) and nuts (52) that hold the air intake pipe (47) to the turbocharger compressor outlet.

Remove the air intake pipe (47).

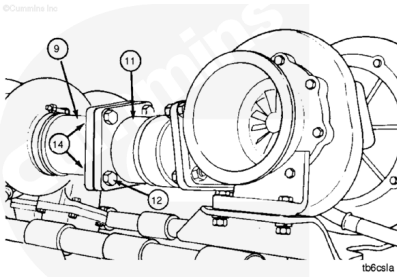
Check the pipe for damage. Discard the part if it is damaged.



Loosen the v-band clamp (8).

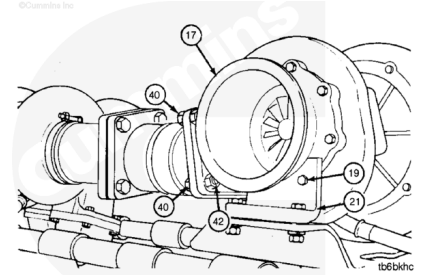


Remove the four capscrews (12) and nuts (14) that hold the two exhaust connections (9 and 11).

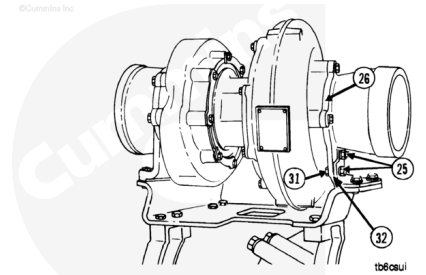


Remove the four capscrews (40) and nuts (42) that hold the exhaust connection to the turbine housing.

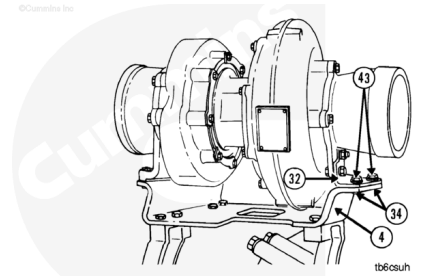
Remove the exhaust connection.



Remove the two capscrews (31) and nuts (25) that hold the bracket (32) to the connection (26).



Remove the two capscrews (43) and the nuts (34) that hold the bracket (32) to the support bracket (4).

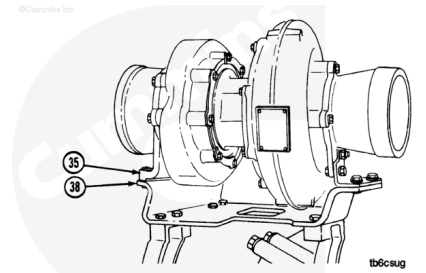


⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Remove the two capscrews (35) and nuts (38) that hold the turbocharger exhaust outlet connection to the bracket.

Remove the turbocharger.



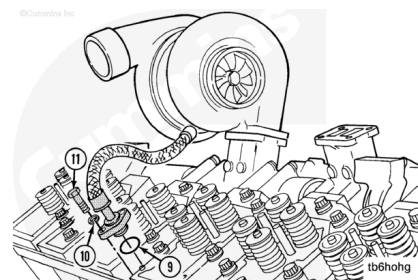
KTТА38, KTTA50, K1500E, K1800E and K2000E High-Pressure, and KT38, KTA38, and KTA50 with Outboard Aftercoolers

If the engine being serviced is a KT38, KTA38, or KTA50, record the direction of the turbocharger exhaust outlet and mark the exhaust manifolds for location. This will aid the assembly process as the exhaust can be either front out or rear out.

Detach the oil drain hose from the cylinder block. Remove the o-ring seal (9).

Discard the o-ring seal.

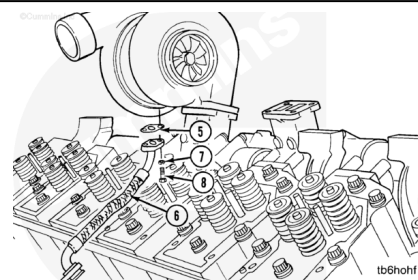
Check the hose for damage. Discard the part if it is damaged.



Detach the oil drain hose from the turbocharger. Remove any support clamps and the hose. Remove the gasket (5).

Discard the gasket.

Check the hose for damage. Discard the part if it is damaged.

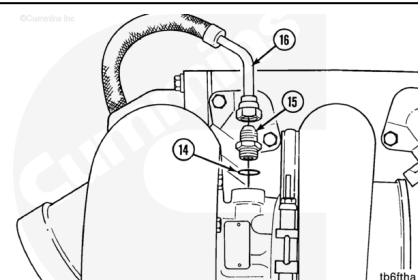


Removal of the fitting (15) and the o-ring (14) is **not** required.

Detach the oil inlet hose. Remove any support clamps.

Remove the hose.

Check the hose for damage. Discard the part if it is damaged.

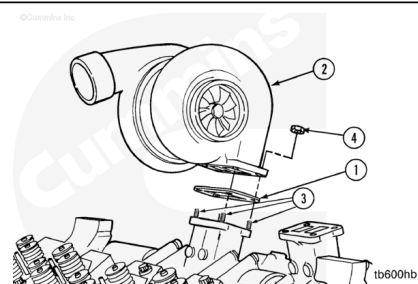


⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

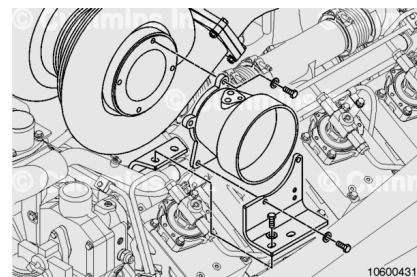
Remove the four capscrews (3) and the nuts (4). Remove the turbocharger and gasket.

Discard the gasket.

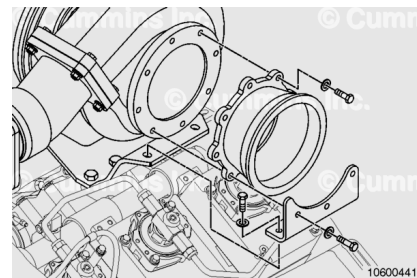


with Electronically Actuated Injector

Remove the low-pressure turbocharger air intake and bracket.

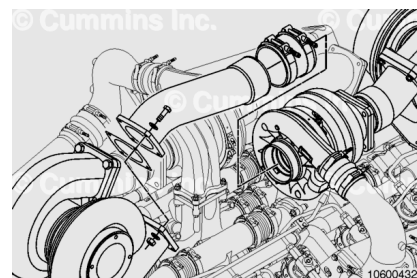


Remove the low-pressure turbocharger exhaust adapter and bracket.



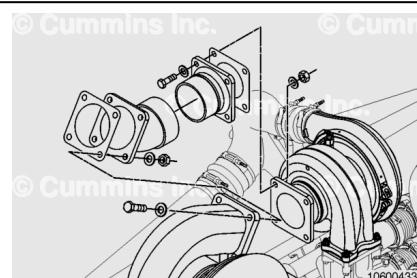
Remove the eight nuts, plain washers, nuts, capscrews and p-clips from the air transfer connections.

Remove the air transfer connections and discard the gaskets.



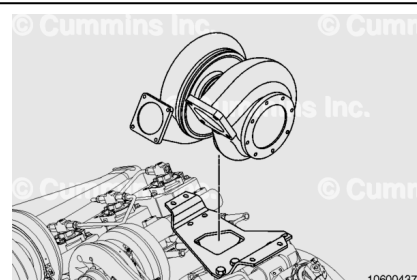
Note : Note the orientation of the capscrews and nuts in the transfer tube prior to removal. The capscrews **must** be installed in the same orientation during installation.

Remove the eight nuts, plain washers, and capscrews, and remove the air transfer piping. Discard the gaskets.



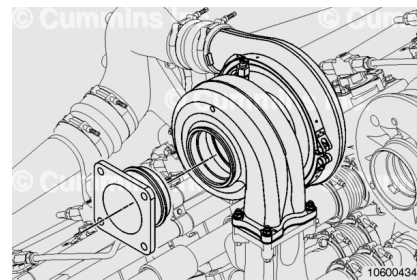
⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



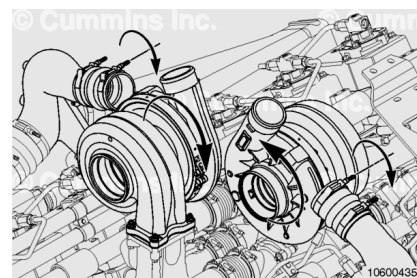
Remove the low-pressure turbochargers.

Loosen and remove the clamps and hoses connecting the turbochargers to the intake manifold.

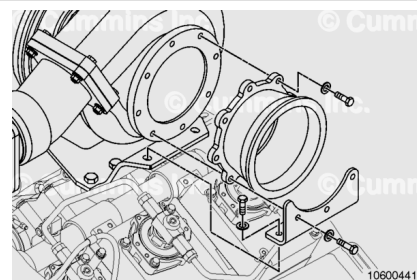


Remove the clamps from the high-pressure turbocharger and air transfer connection.

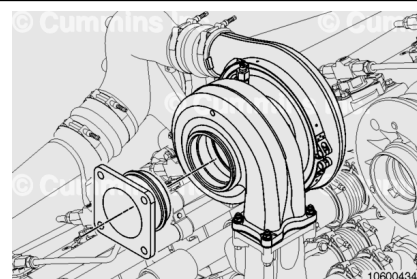
Remove the air transfer from the high-pressure turbocharger.



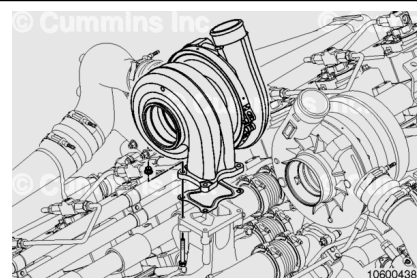
Remove the four capscrews, nuts, and washers and remove the exhaust outlet connection from the high-pressure turbocharger.



Remove the exhaust outlet clamp and outlet connection from the turbocharger.



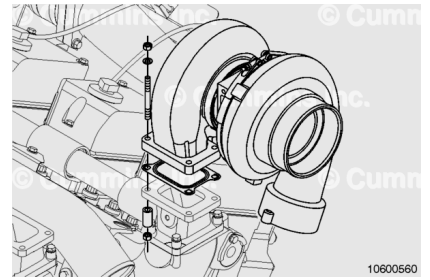
Remove the four mounting capscrews, nuts and washers from each high-pressure turbocharger.



Gas Compression

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the mounting capscrews and lock washers.

Remove the turbocharger and gasket.

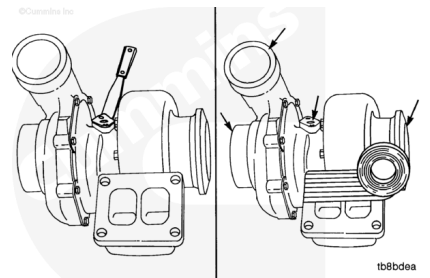
Discard the gasket.

Clean

⚠ CAUTION ⚠

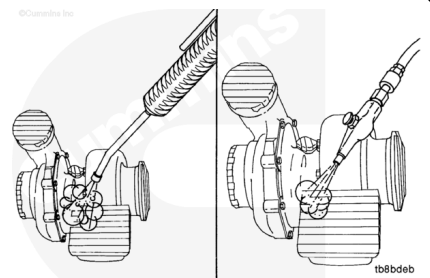
Tape or plug all openings to prevent solvent or steam from entering the oil cavities in the turbocharger.

This procedure applies to all models of turbocharger on K38 and K50 engines. See General Information at the beginning of this section.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



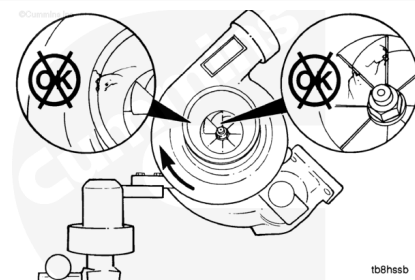
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

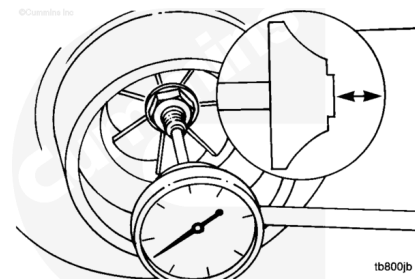
Use solvent or steam. Clean the exterior of the turbocharger.

Inspect for Reuse

Inspect the turbine and compressor wheels for fretting and broken vanes.



Use a dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).

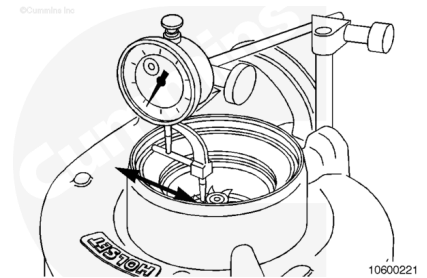


Compressor Impeller Axial End Clearance			
	mm		in
Holset® HC5, HC5A, (127 mm wheel)	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HC5, HC5A, (130 mm wheel)	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX80/HX8 5	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX82/HX8 3	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HE800	0.035	MIN	0.001
	0.115	MAX	0.005
Holset® HT100	0.050	MIN	0.002
	0.152	MAX	0.006

Make sure the movement is within the MIN/MAX Total Indicator Readings (TIR) values shown above.

Note : If the end clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. Reference the HC5A Turbocharger Shop Manual, Bulletin 3810243, or the HT100 Series Turbocharger Shop Manual, Bulletin 3810386, for rebuild instructions.

Measure the radial clearance (side-to-side) at the compressor nose, using a dial gauge.



Compressor Impeller Radial Clearance

	mm		in
Holset® HC5, HC5A, (127 mm wheel)	0.228	MIN	0.009
	0.686	MAX	0.027
Holset® HC5, HC5A, (130 mm wheel)	0.228	MIN	0.009
	0.762	MAX	0.03
Holset® HX80/HX8 5	0.254	MIN	0.01
	0.787	MAX	0.031
Holset® HX82/HX8 3	0.254	MIN	0.010
	0.787	MAX	0.031
Holset® HE800	0.254	MIN	0.010
	0.787	MAX	0.031
Holset® HT100	0.254	MIN	0.01
	0.813	MAX	0.032

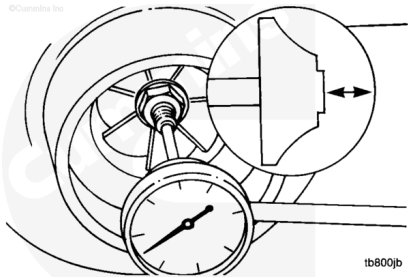
Make sure the movement is within the MIN/MAX TIR values shown above.

Note : If the clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. See HC5A Turbo Shop Manual, Bulletin 3810243, or HT100 Series

Turbocharger Shop Manual, Bulletin 3810386, for rebuild instructions.

Note : Check the model of turbocharger installed on the engine, this is for: BrownBoveri®, Model RR-153 and RR-151.

To measure axial clearance, use Part Number ST-537, Dial Depth Gauge, or a dial indicator. Measure the axial motion.



BBC (RR-151) and (RR-153) End Clearance

mm		in
0.09	MIN	0.004
0.13	MAX	0.005

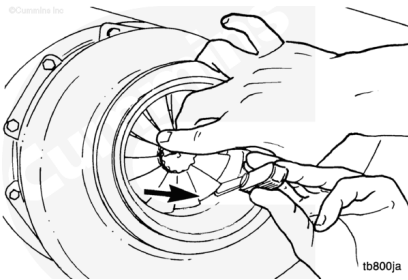
If the end clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Use the following manual for rebuild instructions. Reference the BrownBoveri® Turbocharger Manual, Bulletin 3810235.

Note : The compressor and turbine casings do not require removal to measure the radial clearance. The parts shown are removed for clarity.

Use a feeler gauge to measure the radial clearance.

BBC (RR-151) and (RR-153) Bearing Radial Clearance

	mm		in	
RR-153		0.55	MAX	0.021
RR-151		0.75	MAX	0.030



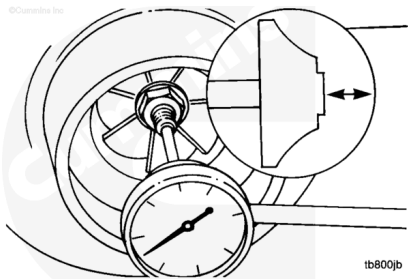
If the radial clearance exceeds the specification, the part **must** be replaced or rebuilt. Use the following manual for rebuild instructions. Reference the BrownBoveri® Turbocharger Manual, Bulletin 3810235.

Note : Check the model of turbocharger installed on the engine, this is for: AiResearch™, Model T-18A.

Use a dial indicator to measure the end clearance.

AiResearch™ T-18A End Clearance

mm		in
0.10	MIN	0.004



AiResearch™ T-18A End Clearance

mm		in
0.23	MAX	0.009

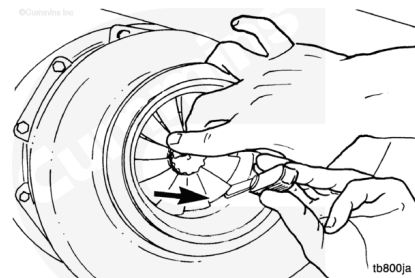
If the end clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Use the following manual for information concerning rebuild instructions. Reference the Turbocharger Component Shop Manual, Bulletin 3379091.

AiResearch® Method One:

Check the radial clearance of both the compressor impeller and turbine wheel.

Use your hand to push the compressor impeller or turbine wheel toward the housing.

Use a feeler gauge that is **not** more than 13 mm [0.5 in] wide. Measure the clearance between the impeller and the housing at the minimum clearance point.



AiResearch® T-18A Impeller and Turbine Wheel Radial Clearance

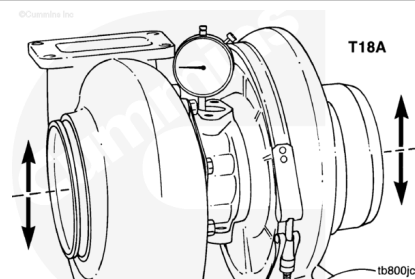
mm		in
0.08	MIN	0.003
0.18	MAX	0.007

If the clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Use the following manual for information concerning rebuild instructions. Reference the Turbocharger Component Shop Manual, Bulletin 3379091.

AiResearch® Method Two:

To measure the direct bearing radial clearance on the T-18A, place a dial indicator through the oil drain hole as illustrated. Be sure the indicator touches the highest part of the shaft.

Move the turbine and compressor wheels by hand to the upper and lower limits.



AiResearch® T-18A Bearing Radial Clearance

mm		in
0.20	MIN	0.008
0.46	MAX	0.018

If the clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. Use the following manual for information concerning rebuild instructions. Reference the Turbocharger Component Shop Manual, Bulletin 3379091.

All models:

After the turbocharger has been checked:

- Lubricate the bearings with engine oil through the oil inlet fitting
- Cover all openings to prevent entry of dirt until installed on the engine.

Note : Check the model of the turbocharger installed on the engine. This is for: Honeywell®, all models.

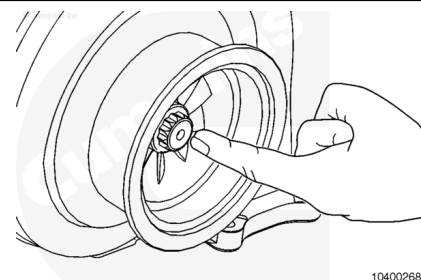
Specifications for checking radial and axial clearance are **not** available.

Inspect for signs of contact between the turbocharger vanes and housing.

Check the bearing system for wear:

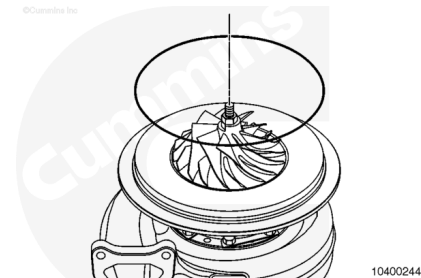
- Hold both ends of the turbocharger shaft and rotate the shaft while applying pressure down on the compressor side and up on the turbine side. It should rotate freely and there should be **no** contact between the compressor wheel and the compressor housing, or the turbine wheel and the turbine housing.

Replace the turbocharger if any malfunctions are identified.

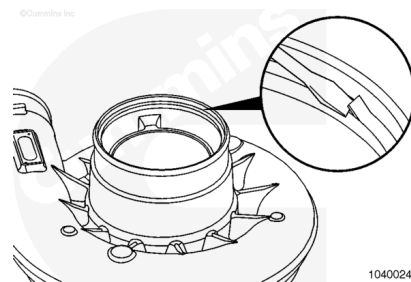


Disassemble

Remove the o-ring and discard.



If the compressor cover has an inlet baffle it can be removed with a flat tip screwdriver. Pry the retaining ring up and remove the retaining ring.

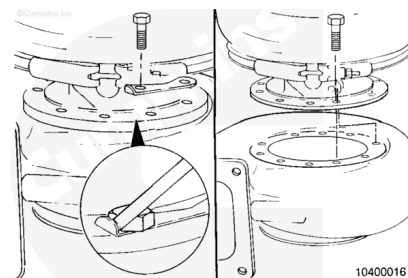


⚠ CAUTION ⚠

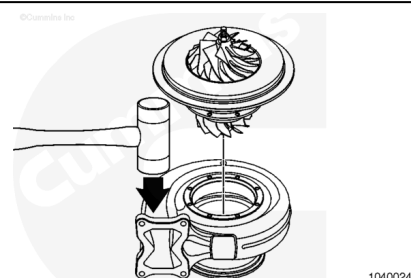
Use caution in handling the turbine blades. The turbine blades can be easily damaged when the turbine housing is removed.

Remove the turbine housing by bending the lock plate tabs away from the capscrew heads.

Remove the capscrews.

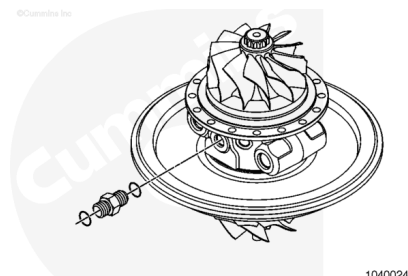


Make sure the turbine wheel rotates freely inside the turbine housing. Tap on the turbine housing with a mallet to separate the turbine housing from the bearing housing.



Remove the oil supply fittings and the coolant lines from the center housing assembly.

Discard the o-rings.



Clean and Inspect for Reuse

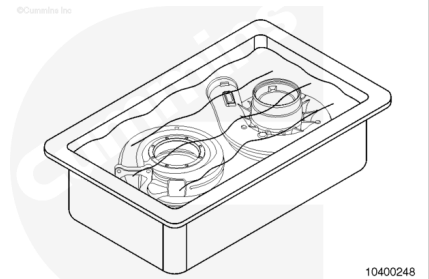
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

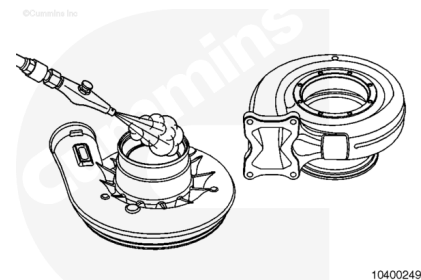
Clean the compressor cover and turbine housing in parts cleaner to loosen deposits.



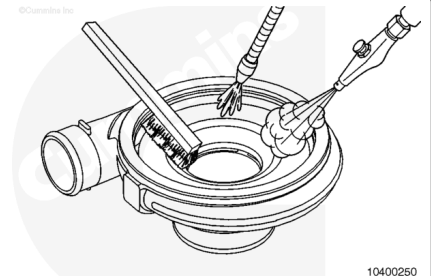
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Dry the compressor housing with compressed air.



Use a soft bristle brush to remove heavy scale deposits.
Wash and dry the parts after brushing.



⚠ CAUTION ⚠

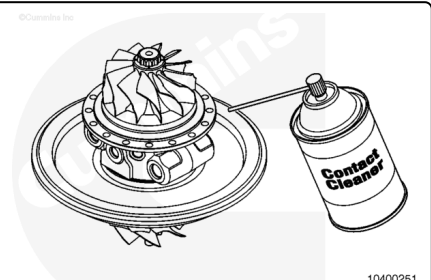
Do not use compressed air to dry or spin the turbine wheel as this can cause damage to the turbocharger bearing system.

If chemical and brush cleaning is **not** effective, bead blast with plastic abrasive media, Part Number 3822735, or equivalent.

Wash and dry parts after bead blasting.

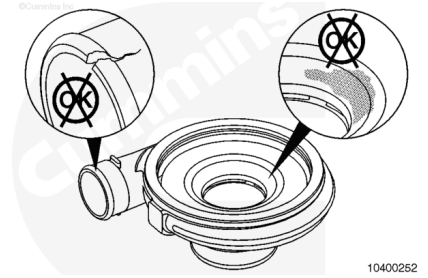
The center housing rotating assembly can be cleaned with electrical contact cleaner, Part Number 3824510, or equivalent.

Do **not** spray contact cleaner into the oil and coolant ports.



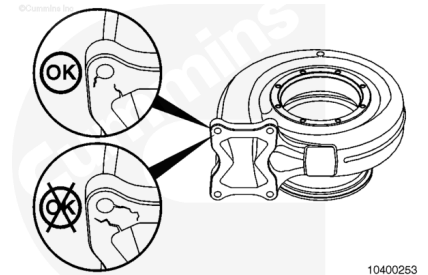
Inspect the compressor cover for cracks, dents, and other damage.

Inspect the inside of the compressor cover for signs of scoring or damage from the compressor wheel. Replace the compressor cover, if damaged.

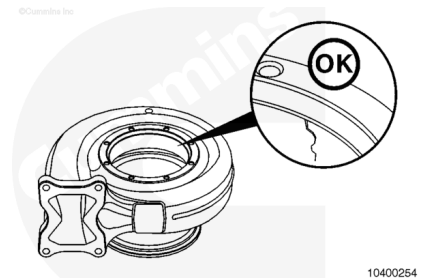


Inspect the turbine housing for cracks and damage. Cracks larger than 10 mm [0.39 in] can cause an exhaust leak. Replace the turbine housing if damage is found.

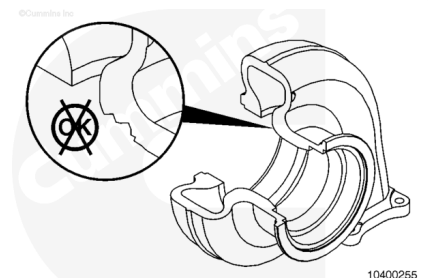
Follow the criteria shown in the illustration.



Minor cracks in the turbine housing internal wall, where exhaust gas flow enters the turbine wheel, is an acceptable condition. The turbine housing can be reused.



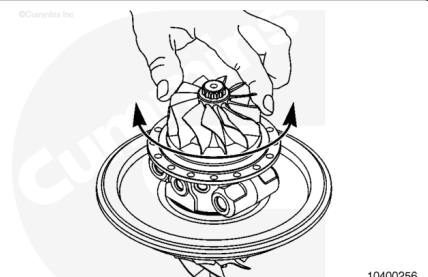
Inspect the turbine housing for signs of contact with the turbine wheel. Damaged turbine housings **must** be replaced.



Inspect the center housing rotating assembly for any signs of cracks, damaged threads, or other damage.

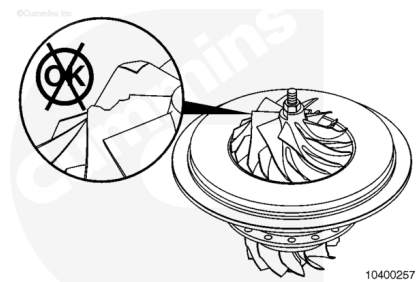
Rotate the turbine wheel by hand to make sure it spins freely.

Replace the center housing rotating assembly if it is damaged or does **not** spin freely.



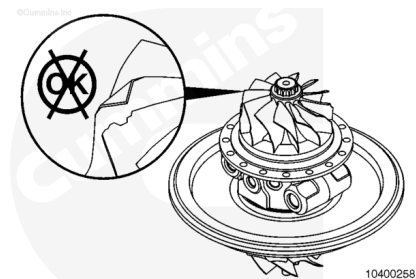
Inspect the compressor wheel for damage, broken, or bent blades.

Do **not** attempt to repair or straighten the blades. Instead, replace the center housing rotating assembly.



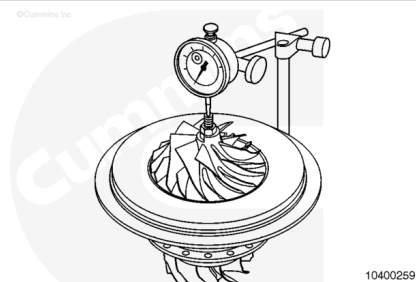
Inspect the turbine wheel for damaged, broken, or bent blades.

Do **not** attempt to repair or straighten the blades. Instead, replace the center housing rotating assembly.



Measure

Use a dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).

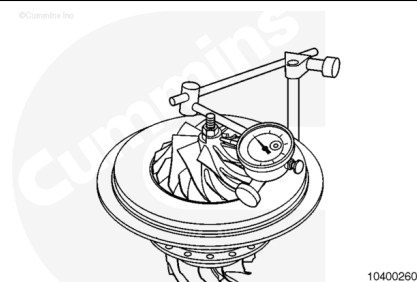


Compressor Impeller Axial End Clearance			
	mm		in
Holset® HC5, HC5A, (127 mm wheel)	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HC5, HC5A, (130 mm wheel)	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX80/HX8 5	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HT100	0.050	MIN	0.002
	0.152	MAX	0.006

Make sure the movement is within the MIN/MAX TIR values shown above.

Replace the center housing rotating assembly if the axial clearance is **not** within specifications.

Measure the radial clearance (side-to-side) at the compressor nose, using a dial gauge.



Compressor Impeller Radial Clearance

	mm		in
Holset® HC5, HC5A, (127 mm wheel)	0.228	MIN	0.009
	0.686	MAX	0.027
Holset® HC5, HC5A, (130 mm wheel)	0.228	MIN	0.009
	0.762	MAX	0.03
Holset® HX80/HX8 5	0.254	MIN	0.01
	0.787	MAX	0.031
Holset® HT100	0.254	MIN	0.01
	0.813	MAX	0.032

Make sure the movement is within the MIN/MAX TIR values shown above.

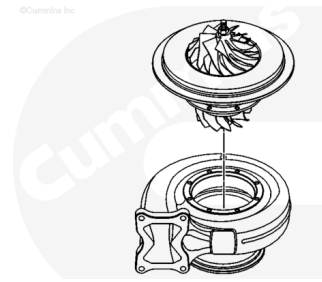
Replace the center housing rotating assembly if the radial clearance is **not** within specifications.

Assemble

⚠ CAUTION ⚠

Use caution when handling the turbine blades. The turbine blades can be easily damaged when installing in the turbine housing.

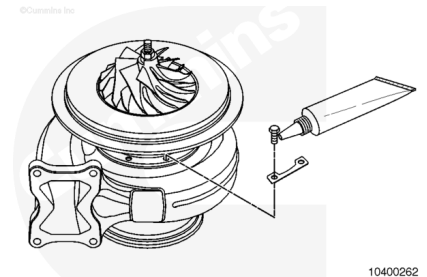
Assemble the center housing rotating assembly on to the turbine housing.



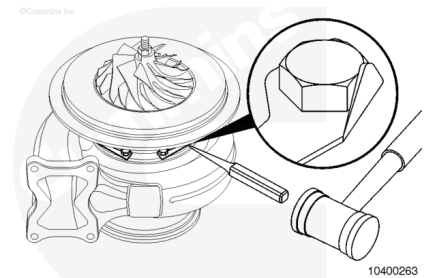
Lubricate the threads of the new capscrews with anti-seize compound, Part Number 3824732, or equivalent.

Install the capscrews, with new lock plates, to the turbocharger turbine housing.

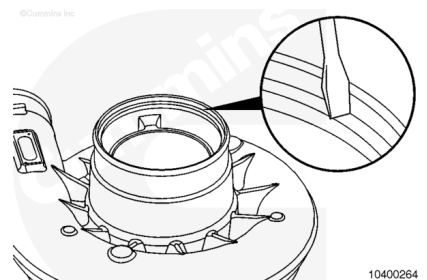
Torque Value: 20.3 n•m [180 in-lb]



Use a hammer and punch to fold the locking plate onto the flat of the capscrew, to prevent loosening.

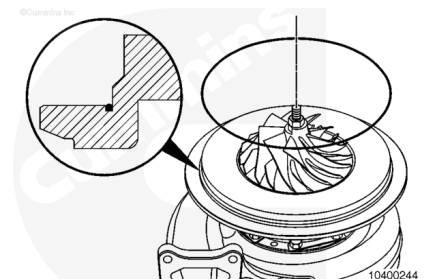


Assemble the inlet baffle into the compressor cover and secure with a retaining ring. Hold one end of the retaining ring in the groove. Press the remainder of the retaining ring into position. Use a flat tip screwdriver to make sure the retaining ring is correctly seated in the compressor cover groove.



Note : The HX83 Turbocharger uses a titanium compressor wheel. The edge of the wheel is very sharp. Handle the compressor wheel with care.

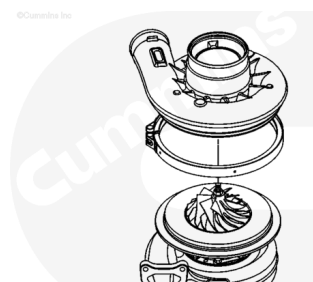
Assemble the compressor housing new o-ring on the turbocharger.



⚠ CAUTION ⚠

Do not damage the compressor wheel blades during assembly.

Assemble the v-band clamp and compressor cover onto the turbocharger.

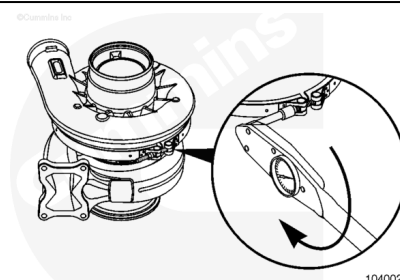


10400266

Place the v-band clamp in position.

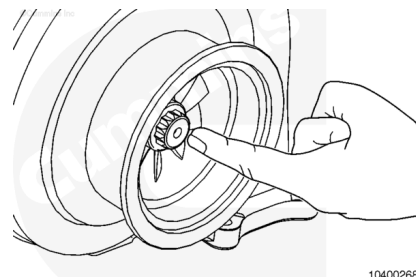
Tighten the v-band clamp.

Torque Value: 12 n•m [106 in-lb]



10400267

Make sure the compressor wheel and turbine wheel rotate freely.



10400268

Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

Check to be sure the correct turbocharger assembly is installed. The turbocharger assembly number is on the dataplate that is mounted on the turbocharger. Compare the assembly number with the number in the Control Parts List (CPL) Manual with the corresponding CPL for the engine being serviced. Many turbochargers appear the same physically, but contain different parts. If the wrong turbocharger assembly is installed, the performance of the engine will be less than specification, or damage can result to the engine due to excessive cylinder pressure and temperature.

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The turbochargers, mounting hardware, and connections are different for various models of K38 and K50 engines. Because of the different parts, installation procedures are given for:

- KT38, KTA38, and KTA50 Turbochargers with Center Mount Aftercooler
- KTTA38 and KTTA50 High-Pressure, and KT38, KTA38, and KTA50 Turbochargers with Outboard Aftercoolers
- KTTA38 Low-Pressure Turbochargers
- KTTA50 Low-Pressure Turbochargers
- K2000E, K1800E and K1500E High-Pressure Turbochargers
- K2000E Low-Pressure Turbochargers

During turbocharger installation, it is often necessary to adjust the angle of the bearing housing oil drain and the angle of the compressor housing outlet. Instructions for these adjustments are described following the procedure for the KTTA50 Low-Pressure Turbocharger.

Oil inlet and drain hose routing and clamps can be different than shown because of the age of the engine. Be sure the oil hoses have the correct clamps installed to prevent damage from vibration or heat from another part.

KT38, KTA38, and KTA50 Turbochargers with Center Mount Aftercooler

Note : The illustrations show a KTA38. The KTA50 has a longer connection between the aftercooler housing and the turbocharger. The procedure is the same for these models.

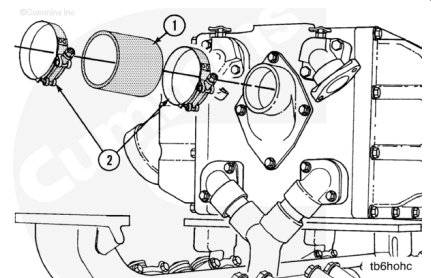
Newer engines have a hose and clamps to connect the turbocharger to the aftercooler connection. Older engines have a tube with a dust seal and two o-ring seals to connect the turbocharger to the aftercooler connection.

The clamps are special to allow a higher torque. Standard T-bolt type clamps can loosen and allow an air leak.

Newer Engines:

Install the hose (1) and clamps (2) on the aftercooler connection. Do **not** tighten the clamps.

Some of the newer K50 engines have a hose and heavy duty clamps installed over the tube and o-ring as shown in the next picture.

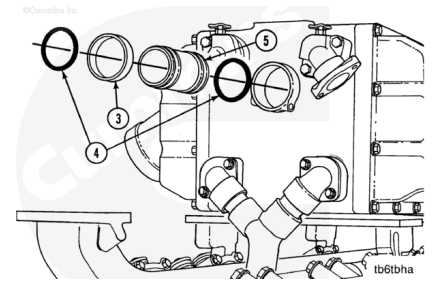


Older Engines:

Install the dust seal (3) and o-ring seals (4) on the tube (5).

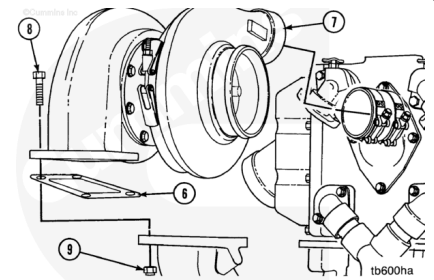
Use vegetable oil. Lubricate the o-ring seals.

Install the tube in the bore of the aftercooler housing. Push the tube into the bore until it touches the step in the bore.



⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the gasket (6) on the exhaust manifold. The gasket **must** be installed so that the raised bead on the gasket is toward the turbocharger.

Install the turbocharger (7). Push the compressor housing into the hose or on the hose.

The capscrews (8) with nuts (9) are manufactured from heat resistant material. If heat resistant capscrews are **not** installed, the parts will fail, causing an exhaust leak.

Apply an anti-seize compound on the threads of the capscrew.

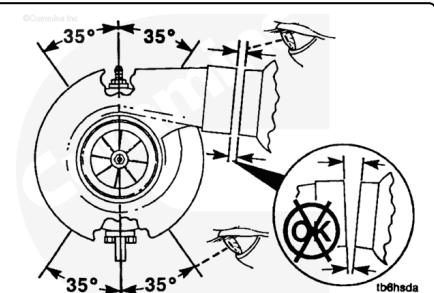
Install the four capscrews (8) with the nuts (9).

Torque Value:

Capscrews 40 n•m [30 ft-lb]

Some of the compressor housing is **not** shown removed for clarity.

Check the angle of the oil drain and the alignment of the compressor housing to the aftercooler connection. If the angle or alignment is **not** correct, adjust the turbocharger oil drain angle, or adjust the turbocharger compressor housing alignment, described later in this section.



Newer Engines:

Tighten the clamps (2).

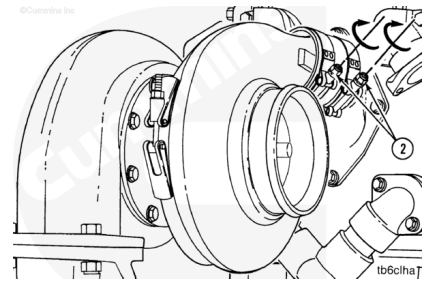
Torque Value:

K38 9 n•m [80 in-lb]

The K50 requires a special clamp.

Torque Value:

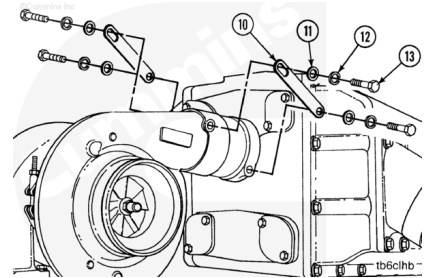
K50 15 n•m [133 in-lb]



Older Engines:

Install the parts.

- Retaining plates (10)
- Flat washers (11)
- Lock washers (12)
- Capscrews (13).

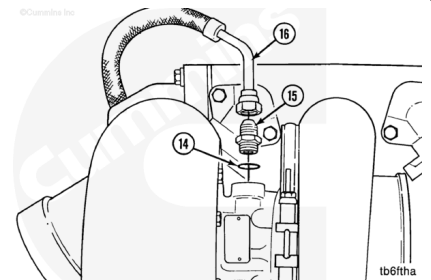


Torque Value:

Capscrews 16 n•m [142 in-lb]

Note : Holset® turbochargers have a fitting that has straight threads and requires an o-ring seal. AiResearch® turbochargers have a fitting with pipe threads. The torque value is the same for both types of the fittings.

Install the o-ring seal (14) on the oil inlet fitting (15). Install the fitting.



Torque Value:

Oil Inlet Fitting 35 n•m [26 ft-lb]

Connect the turbocharger oil inlet hose (16) to the fitting.

Install the clamps for the oil inlet hose.

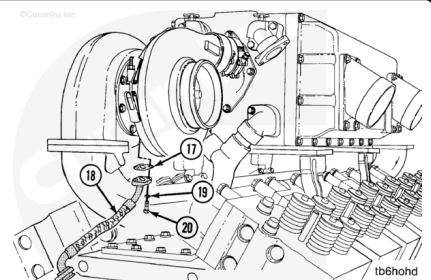
The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html>)

Install the parts.

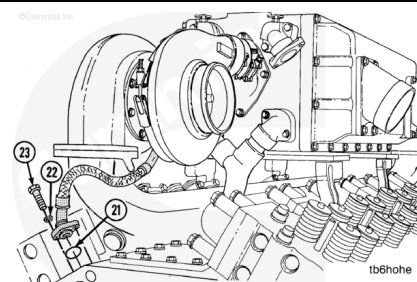
- Gasket (17)
- Hose (18)
- Lock washers (19)
- Capscrews (20).

Torque Value:

Capscrews 40 n•m [30 ft-lb]



Newer engines have a flange and an o-ring seal type oil drain hose as shown. The newer hose connects to the cylinder block with a capscrew. Older engines have a hose that has a tubing nut and connects to the block with a fitting that has pipe threads.



If a flange and an o-ring type connection is installed on an older block, a spacer is required between the hole in the block for the capscrew and the flange. If the spacer is **not** installed, an oil leak will result.

Install the o-ring seal (21) on the hose flange.

Use vegetable oil. Lubricate the seal.

Install the flange, the lock washer (22), and the capscrew (23).

Torque Value:

Flange Capscrews 40 n•m [30 ft-lb]

Repeat this procedure to install the other turbocharger and connect the oil hoses.

KTTA38 and KTTA50 High-Pressure, and KT38, KTA38, and KTA50 Turbochargers with Outboard Aftercoolers

The turbochargers **must** be mounted on the correct engine bank.

KTTA38 and KTTA50 High-Pressure:

The turbocharger on the left bank **must** be installed with the compressor housing pointed toward the front of the engine.

The turbocharger on the right bank **must** be installed with the turbine housing pointed toward the front of the engine.

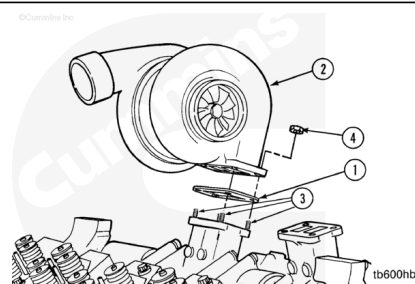
KT38, KTA38, and KTA50 with Outboard Aftercoolers:

The turbine housing on both turbochargers **must** point the same direction, either toward the front of the engine or toward the rear of the engine so that the exhaust piping will fit the application.

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⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The turbochargers for the K50 engine with pulse type exhaust manifolds are to be mounted with four studs and nuts, **not** capscrews and nuts. The torque value is the same. Some K38 engines do **not** require the nuts.

The gasket **must** be installed so that the raised bead on the gasket is toward the turbocharger.

Install the gasket (1) on the exhaust manifold.

Apply an anti-seize compound on the threads of the capscrews or studs (3).

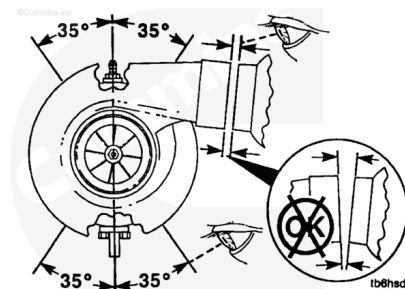
Install the turbocharger (2), the capscrews or studs (3), and the nuts (4), if required.

Torque Value:

Turbocharger Capscrews 40 n•m [30 ft-lb]

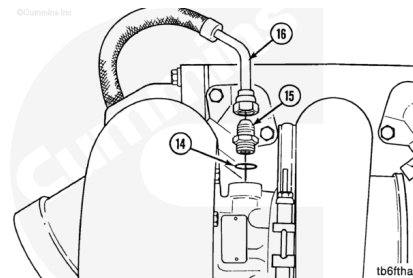
Note : Some of the compressor housing is shown removed for clarity.

Check the angle of the oil drain. The alignment of the compressor housing to the air crossovers can **not** be checked until the aftercooler assembly is installed. If the angle is **not** correct, adjust the turbocharger oil drain angle, described later in this section.



Note : Holset® turbochargers have a fitting that has straight threads and requires an o-ring seal. AiResearch® turbochargers have a fitting with pipe threads. The torque value is the same for both types of fittings.

Install the o-ring seal (14) on the oil inlet fitting (15). Install the fitting.



Torque Value:

Oil Inlet Fitting 35 n•m [26 ft-lb]

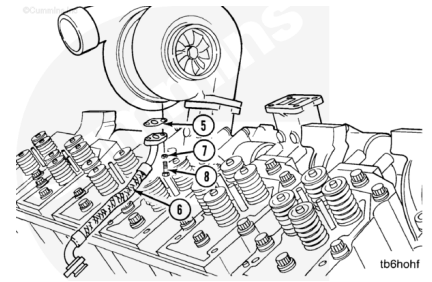
Connect the turbocharger oil inlet hose (16) to the fitting.

Install the clamps for the oil inlet hose.

The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Install the parts.

- Gasket (5)
- Hose (6)
- Lock washers (7)
- Capscrews (8).



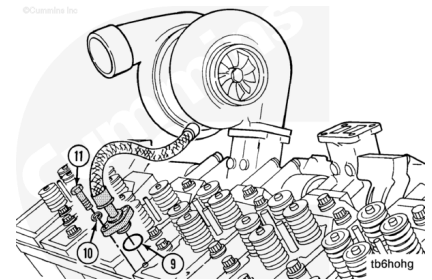
Torque Value:

Capscrews 40 n•m [30 ft-lb]

Install the o-ring seal (9) on the hose flange.

Use vegetable oil. Lubricate the seal.

Install the flange in the bore of the block. Install the lock washer (10) with the cap screw (11).



Torque Value:

Flange Capscrews 40 n•m [30 ft-lb]

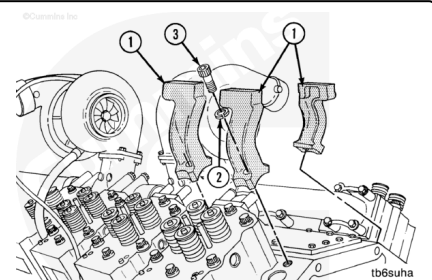
Repeat this procedure to install the other turbocharger and connect the oil hoses.

KTTA38 Low-Pressure Turbochargers

Note : The engine has two low-pressure turbochargers. The illustrations show the installation of one turbocharger. It is recommended that the two low-pressure turbochargers be installed at the same time.

The engine has six turbocharger supports. Two are mounted on the front of the right bank and two on the rear of the left bank. One is mounted on the front of the left bank and one on the rear of the right bank.

Install the turbocharger supports (1) and the lock washers (2), with the socket head capscrews (3).

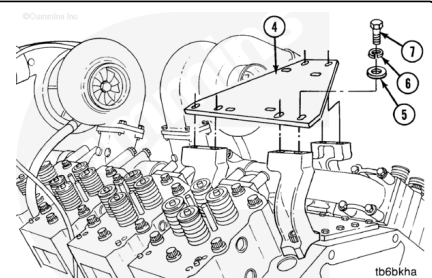


Torque Value:

Turbocharger Support Capscrews 270 n•m [199 ft-lb]

Do **not** tighten the clamps or fasteners to final torque until the torque value is given. Tighten the parts enough to hold them together, but loose enough to allow the parts to be moved for alignment.

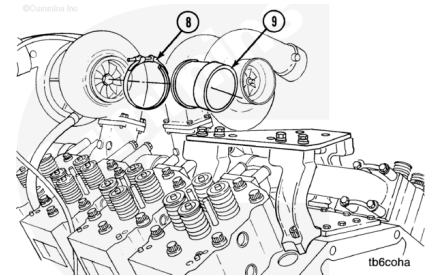
Install the parts.



- Bracket (4)
- Flat washers, heavy (5)
- Lock washers (6)
- Capscrews (7).

There are six each of the flat washers, lock washers, and capscrews.

Install the v-band clamp (8) and the exhaust outlet connection (9).



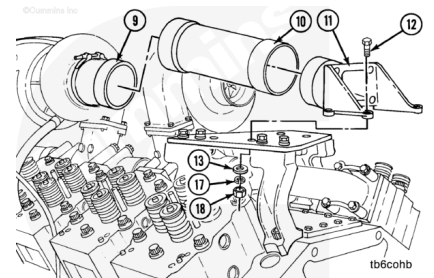
Apply an anti-seize compound to the slip-joints of the exhaust outlet connection (9), exhaust outlet pipe (10), and the turbocharger connection (11).

Install the pipe (10) to the connection (9). Install the connection (11) in the pipe (10).

Apply an anti-seize compound on the threads of the capscrews (12).

Install the four capscrews (12) through the connection and bracket.

Install the heavy flat washers (13), the lock washers (17), and the nuts (18).

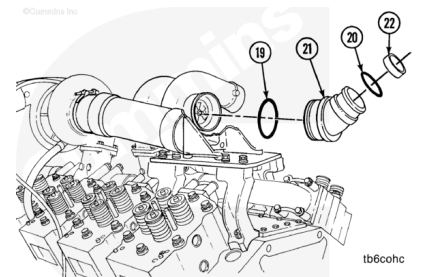


Install the large o-ring seal (19) and the small o-ring seal (20) on the air inlet pipe (21).

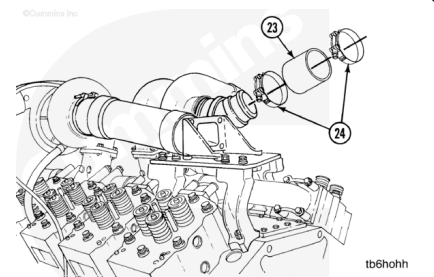
Install the dust seal (22) on the air inlet pipe.

Use vegetable oil. Lubricate the o-ring seals.

Install the air inlet pipe in the high-pressure turbocharger.

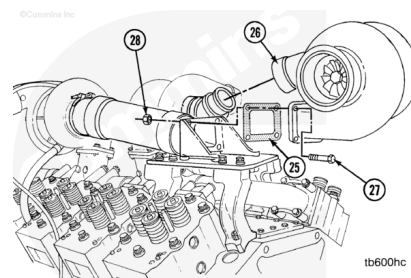


Install the hose (23) and the two T-bolt clamps (24) on the air inlet pipe and high-pressure turbocharger.



⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The gasket **must** be installed so that the raised bead on the gasket is toward the turbocharger

Install the gasket (25).

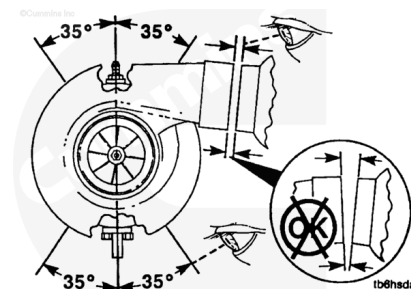
Install the low-pressure turbocharger (26). Push the compressor housing on the air inlet pipe.

Apply an anti-seize compound to the threads of the capscrews (27).

Install the four capscrews (27) and lock nuts (28).

Note : Some of the compressor housing is shown removed for clarity.

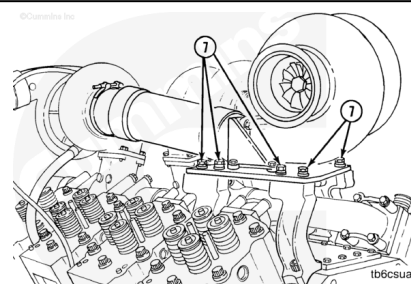
Check the angle of the oil drain. Check the alignment of the compressor housing to the air inlet pipe. If the angle or alignment is **not** correct, see Turbocharger, Oil Drain Angle, Adjust, or Turbocharger, Compressor Housing Alignment, Adjust, described in this section.



Tighten the six capscrews (7) that hold the bracket to the supports.

Torque Value:

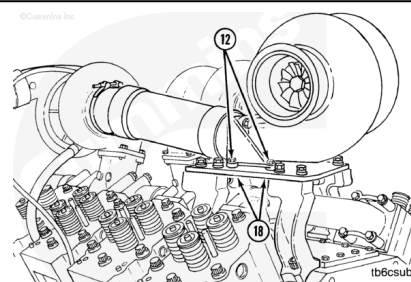
Bracket to Supports 90 n·m [66 ft-lb]



Tighten the four capscrews (12) and nuts (18) that hold the connection to the bracket.

Torque Value:

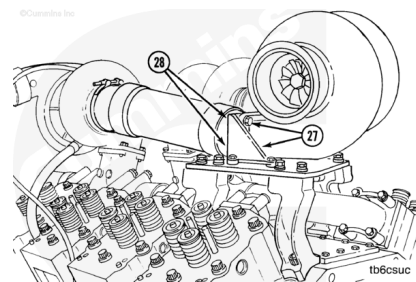
Connection to Bracket 90 n·m [66 ft-lb]



Tighten the four capscrews (27) and nuts (28) that hold the turbocharger to the connection.

Torque Value:

Turbocharger Connection 40 n•m [30 ft-lb]



Tighten the v-band clamp (8).

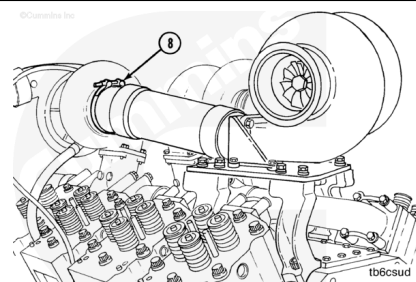
Torque Value:

V-band Clamp 7 n•m [62 in-lb]

Use a mallet. Tap around the clamp to be sure it is seated.

Torque the clamp nut to the same torque value.

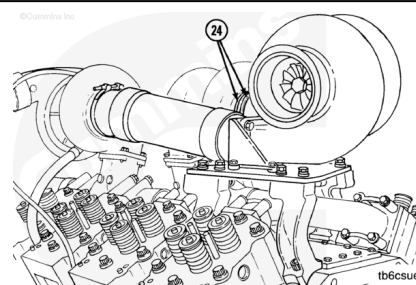
Continue tapping and tightening the clamp until the nut does **not** turn at the specified torque after tapping.



Tighten the two T-bolt clamps (24).

Torque Value:

T-bolt Clamps 9 n•m [80 in-lb]



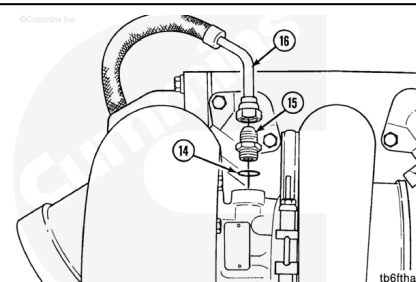
Note : Holset® turbochargers have a fitting that has straight threads and requires an o-ring seal. AiResearch® turbochargers have a fitting with pipe threads. The torque value is the same for both types of fittings.

Install the o-ring seal (14) on the oil inlet fitting (15). Install the fitting.

Torque Value:

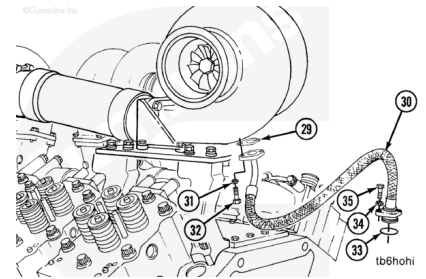
Oil Inlet Fitting 35 n•m [26 ft-lb]

The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)



Install the following parts.

- Gasket (29)
- Hose (30)
- Lock washers (31)
- Capscrews (32).



Torque Value:

Capscrews 40 n•m [30 ft-lb]

Note : Some older KTTA38 engines equipped with AiResearch® low-pressure turbochargers have a drain hose that has two pieces connected by a temperature resistant hump hose.

Install the o-ring seal (33) on the hose flange.

Use vegetable oil. Lubricate the seal.

Install the flange, the lock washer (34), and the capscrew (35).

Torque Value:

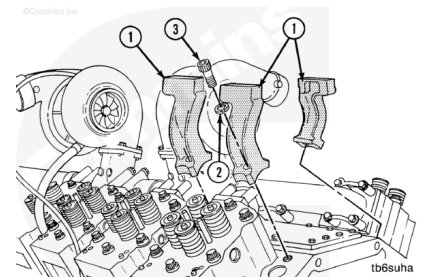
Flange Capscrews 40 n•m [30 ft-lb]

KTTA50 Low-Pressure Turbochargers

Note : The engine has two low-pressure turbochargers. The illustrations show the installation of one turbocharger. It is recommended that the two low-pressure turbochargers be installed at the same time.

The engine has six turbocharger supports. Two are mounted on the front of the right bank, and two are on the rear of the left bank. One is mounted on the front of the left bank, and one on the rear of the right bank.

Install the turbocharger supports (1), the lock washers (2), and the socket head capscrews (3).

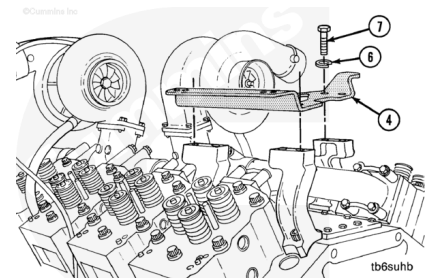


Torque Value:

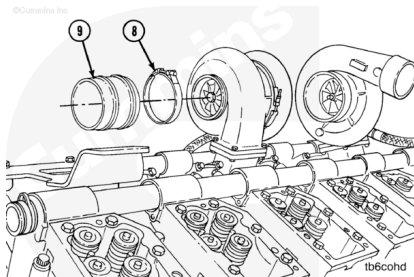
Turbocharger Support Capscrews 270 n•m [199 ft-lb]

Do **not** tighten the clamps or fasteners to final torque until the torque value is given. Tighten the parts enough to hold them together, but loose enough to allow the parts to be moved for alignment.

Install the bracket (4), the six lock washers (6), and the six capscrews (7).

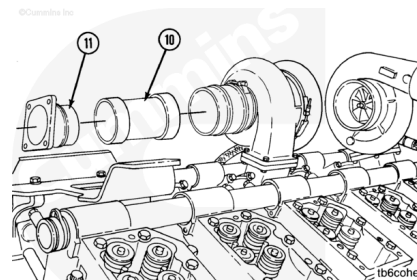


Install the v-band clamp (8) and the exhaust outlet connection (9).



Apply an anti-seize compound to the slip-joints of the exhaust outlet connection (9), the exhaust outlet pipe (10), and the turbocharger connection (11).

Install the pipe (10) on the outlet connection (9). Install the connection (11) in the pipe (10).

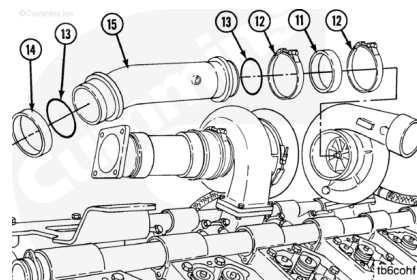


Install the hose (11) and the two T-bolt clamps (12) on the high-pressure turbocharger.

Install the two o-ring seals (13) and the dust seal (14) on the air inlet pipe (15).

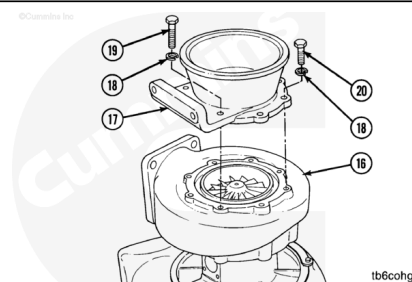
Use vegetable oil. Lubricate the o-ring seals.

Install the air inlet pipe in the high-pressure turbocharger.



⚠ CAUTION ⚠

KTТА50 engines equipped with LOG type exhaust manifolds must have Brown-Boveri Corporation (BBC®), Model Number 153, turbochargers. KTTA50 engines with PULSE type exhaust manifolds must have BBC®, Model Number RR 151, turbochargers. Be sure the correct turbochargers are installed, or damage to the engine can result.



⚠ CAUTION ⚠

Newer BBC® RR 153 turbochargers have fastener threads that are U.S. Customary inch. Older BBC® RR 153 turbochargers have fastener threads that are metric millimeter. The fastener threads in all BBC® RR 151 are metric millimeter. Be sure the correct fasteners are installed or the parts can be damaged.

Note : The exhaust and intake adapters for the BBC® RR 153 and RR 151 appear the same, but have different dimensions. Be sure to install the correct part.

Apply an anti-seize compound to the bore of the low-pressure turbocharger (16) for the exhaust outlet connection (17), and the threads of the capscrews (19) and (20).

The connection **must** be installed as shown so that the flange with two holes aligns with the bracket.

The two capscrews (19) are longer than the six capscrews (20).

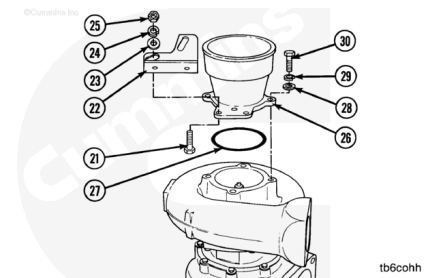
Install the exhaust outlet connection (17), the lock washers (18), and the capscrews (19) and (20).

Torque Value:

Exhaust Outlet Connection Capscrews 40 n•m [30 ft-lb]

Install the parts on the air intake connection (26).

- Capscrews (21)
- Bracket (22)
- Flat washers, heavy (23)
- Lock washers (24)
- Nuts (25).



Install the o-ring seal (27) in the groove on the connection (26).

Use engine oil or lubriplate to secure the seal in the groove.

Install the parts on the low-pressure turbocharger.

- Connection (26)
- Flat washers (28)
- Lock washers (29)
- Capscrews (30).

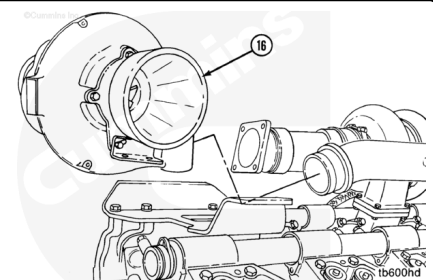
Tighten **only** the capscrews (30).

Torque Value:

Capscrews 25 n•m [18 ft-lb]

⚠ WARNING ⚠

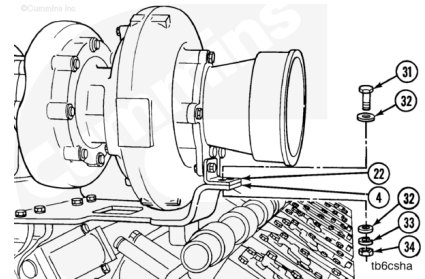
The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a hoist and lifting strap. Put the turbocharger (16) in position and push the compressor housing on the air pipe.

Install the capscrews (31) through the heavy flat washers (32), and the brackets (22) and (4).

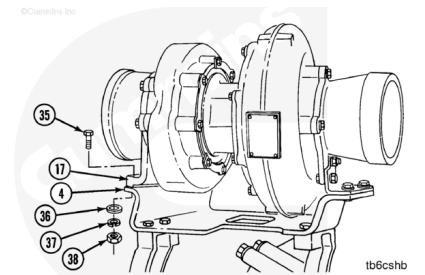
Install the heavy flat washers (32), lock washers (33), and the nuts (34).



Apply an anti-seize compound to the threads of the capscrews (35).

Install the two capscrews (35) through the exhaust outlet connection (17) and the bracket (4).

Install the heavy flat washers (36), the lock washers (37), and the nuts (38).



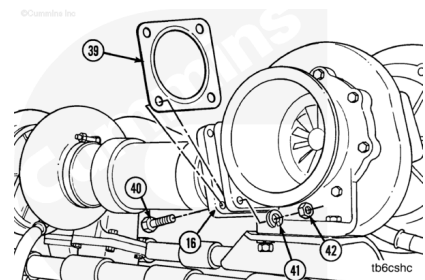
The gasket **must** be installed so that the raised bead on the gasket is toward the turbocharger.

Install the gasket (39) between the turbocharger turbine housing and the exhaust connection.

Apply an anti-seize compound to the threads of the capscrews (40).

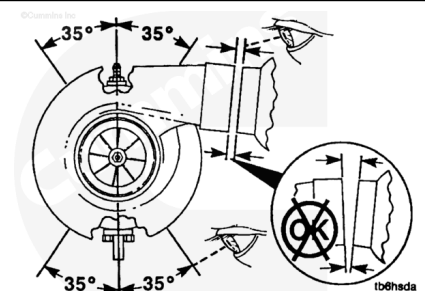
Install the four capscrews (40) through the holes in the turbine housing, the gasket, and the connection.

Install the lock washers (41) and the nuts (42).



Note : Some of the compressor housing is shown removed for clarity.

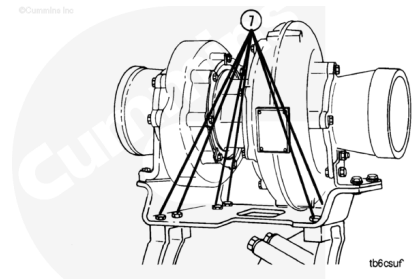
Check the angle of the oil drain. Check the alignment of the compressor housing to the air inlet pipe. If the angle or alignment is **not** correct, see Turbocharger, Oil Drain Angle, Adjust, or Turbocharger, Compressor Housing Alignment, Adjust, described in this section.



Tighten the six capscrews (7) that hold the bracket to the supports.

Torque Value:

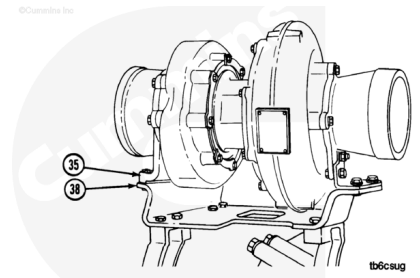
Bracket to Supports 90 n•m [66 ft-lb]



Tighten the two capscrews (35) and nuts (36) that hold the turbocharger exhaust outlet connection to the bracket.

Torque Value:

Exhaust Outlet Connection to Bracket 90 n•m [66 ft-lb]

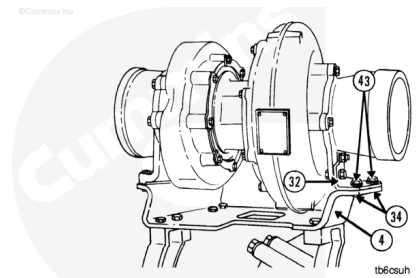


Note : Be sure the bracket (32) is flat against the flange on the air inlet connection.

Tighten the two capscrews (43) and the nuts (34) that hold the air intake connection (32) to the bracket (4).

Torque Value:

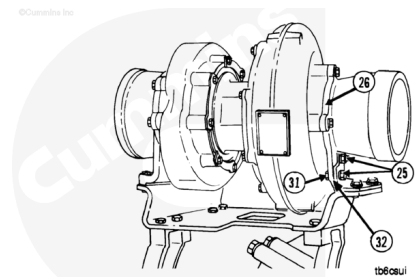
Air Intake Connection to Bracket 40 n•m [30 ft-lb]



Tighten the two capscrews (31) and the nuts (25) that hold the bracket (32) to the connection (26).

Torque Value:

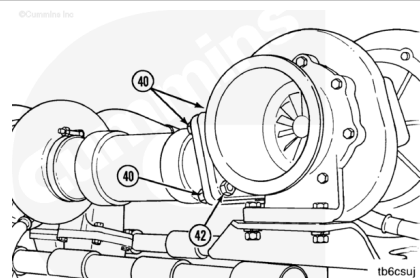
Bracket Capscrews 40 n•m [30 ft-lb]



Tighten the four capscrews (40) and nuts (42) that hold the exhaust connection to the turbine housing.

Torque Value:

Exhaust Connection Capscrews 90 n•m [66 ft-lb]



Tighten the v-band clamp (8).

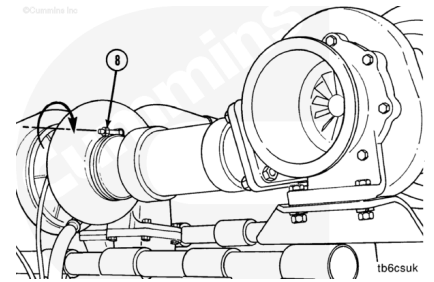
Torque Value:

V-band Clamp 7 n•m [62 in-lb]

Use a mallet. Tap around the clamp to be sure it is seated.

Tighten the clamp nut to the same torque value.

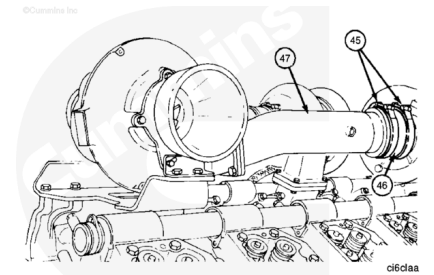
Continue tapping and tightening the clamp until the nut does **not** turn at the specified torque after tapping.



Tighten the two T-bolt clamps (45) on the air inlet pipe.

Torque Value:

T-bolt Clamps 7 n•m [62 in-lb]



⚠ CAUTION ⚠

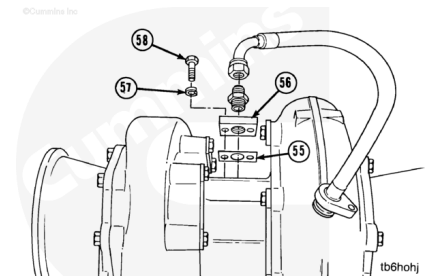
Be sure the correct capscrews are installed. If the wrong capscrews are installed, failure of the part can result.

Note : Newer BBC® RR153 turbochargers have a fitting that is threaded directly into the bearing housing. Older BBC® RR 153 turbochargers require an adapter plate, Part Number 3176567, and a gasket, Part Number 3176568, attached to the bearing housing with two capscrews.

Note : All BBC® RR 151 turbochargers have an adapter plate and gasket. The adapter plate and gasket are different than on the older BBC® RR 153 turbochargers.

Install the parts for the turbocharger oil inlet.

- Gasket (55)
- Plate (56)
- Lock washers (57)
- Capscrews (58).



Torque Value:

Oil Inlet Capscrew 40 n•m [30 ft-lb]

If the fitting is **not** installed in the plate, install the oil inlet fitting.

Torque Value:

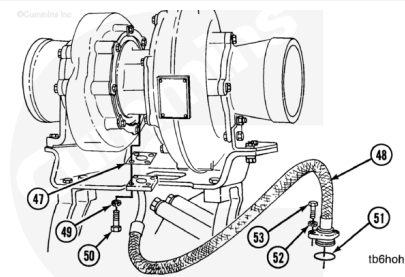
Fitting Torque 35 n•m [26 ft-lb]

Connect the turbocharger oil inlet hose.

The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Install the parts for the turbocharger oil drain.

- Gasket (47)
- Hose (48)
- Lock washers (49)
- Capscrews (50).



Torque Value:

Oil Drain Capscrews 40 n•m [30 ft-lb]

Install the o-ring seal (51) on the hose flange.

Use vegetable oil. Lubricate the seal.

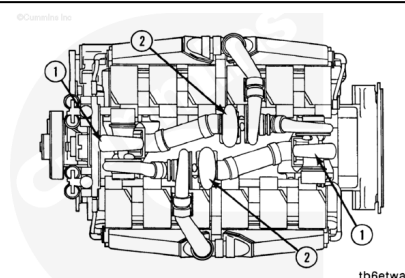
Install the flange, the lock washer (52), and the capscrews (53) in the proper holes in the cylinder block.

Torque Value:

Oil Drain Flange Capscrews 40 n•m [30 ft-lb]

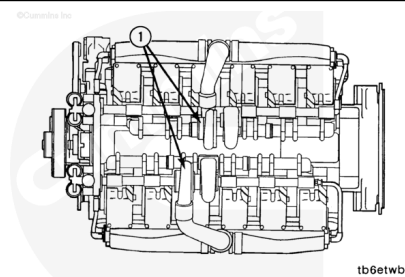
K2000E and K1800E High-Pressure Turbochargers

The K2000E uses Holset® HC5A High-Pressure Turbochargers (2), the HT100 Low-Pressure Turbochargers (1) installation is described later in this section.

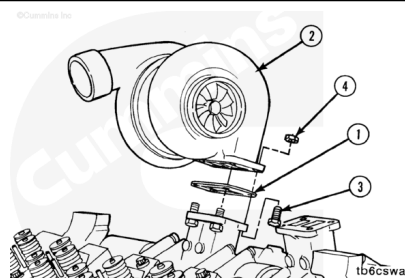


The K1800E engine is single stage turbocharged and does **not** use the turbocharger interstage connections or assembly procedures as does the K2000E.

The K1800E engine uses Holset® HX85 turbochargers (1).

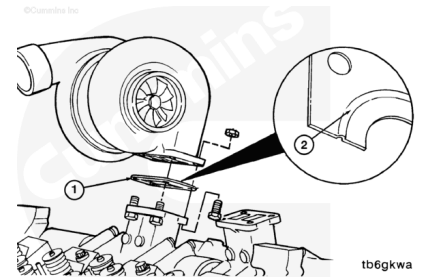


Note : The high-pressure turbochargers (2) are mounted to the exhaust manifold using four capscrews (3) and nuts (4). These capscrews and nuts are manufactured from high strength stainless steel that has high temperature capability. Do **not** substitute with lower quality parts.



The gasket **must** be installed with the raised bead (2) toward the turbocharger.

Install the gasket (1) on the exhaust manifold.

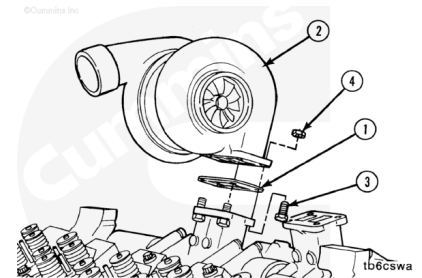


Coat the threads of the capscrews with anti-seize compound.

Install the turbocharger (2), capscrews (3) and nuts (4).

Torque Value:

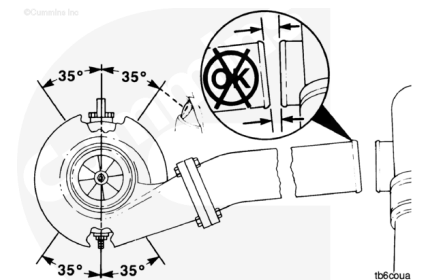
Turbocharger Mounting Capscrews 40 n•m [30 ft-lb]



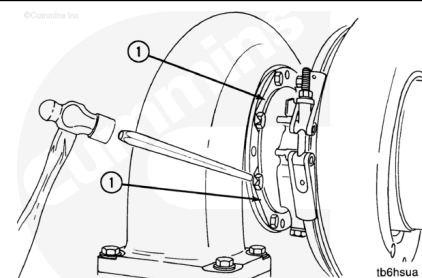
Note : Some of the compressor housing has been removed for clarity.

Check the angle of the oil drain and the alignment of the compressor housing to the air crossovers (this can **not** be checked until the aftercooler assemblies are installed).

If the angle is **not** correct, the bearing housing **must** be rotated in relation to the turbine housing. Adjust the turbocharger oil drain angle.



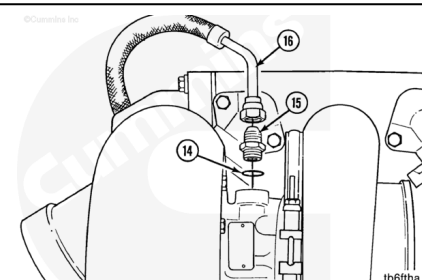
If the alignment is correct or after the adjustment, lock the bearing housing to the turbine casing lockplate tabs.



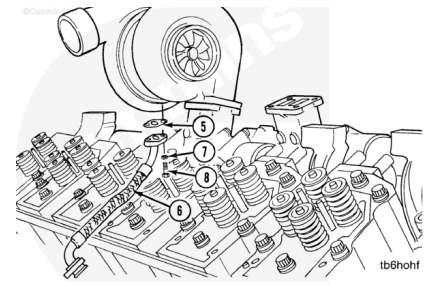
Connect the turbocharger oil inlet hose (16) to the fitting (15).

Install the clamps for the oil inlet hose.

The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html>)



Install the gasket (5), hose (6), lock washers (7) and two capscrews (8) to the turbocharger oil drain.

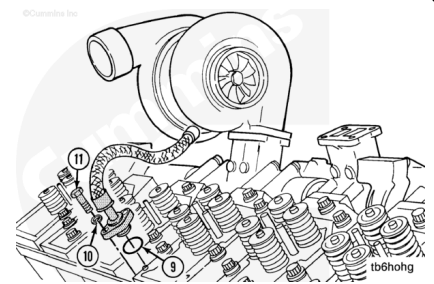


Install the o-ring seal (9) on the hose flange. Use engine oil to lubricate the seal. Install the flange in the bore of the block. Install the lock washer (10) and the capscrew (11).

Torque Value:

Oil Inlet Capscrews 40 n•m [30 ft-lb]

Repeat the procedure to install the other turbocharger and connect the oil supply and the drain hose.

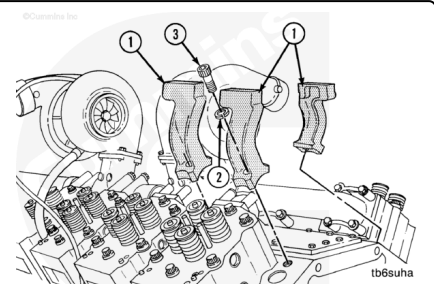


K2000E Low-Pressure Turbochargers

The K2000E engine contains two low-pressure turbochargers. This illustration shows the installation of **only** one turbocharger. The recommendation is to install both low-pressure turbochargers at the same time.

The K2000E engine contains six turbocharger supports. Two of each of the supports are mounted on the front of the right bank and rear of the left bank. One of each of the supports is mounted on the front of the left bank and rear of the right bank.

Install the turbocharger supports (1), and the lock washers (2), and the socket head capscrews (3).

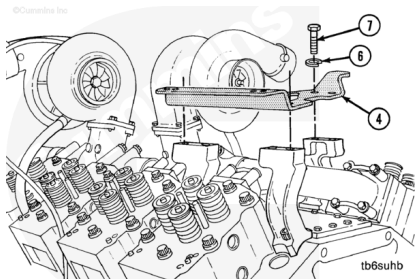


Torque Value:

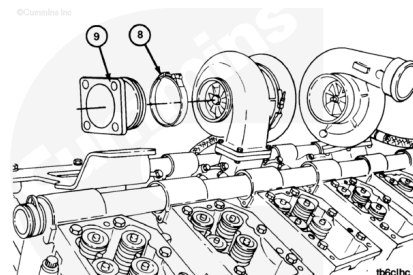
Turbocharger Support Capscrews 270 n•m [199 ft-lb]

Note : From this point on, do not tighten the clamps or fasteners to their final torque unless the torque value is given. Tighten **only** enough to allow the parts to move for alignment.

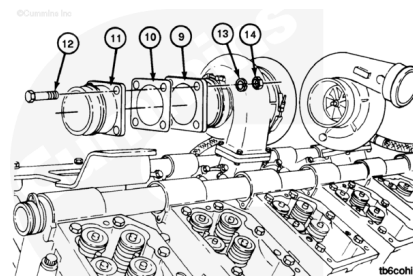
Install the bracket (4), six lock washers (6), and six capscrews (7).



Install the v-band clamp (8) and exhaust outlet connection (9).



Apply anti-seize compound to the threads of the capscrews (12). Install the gasket (10), capscrews (12), lock washers (13), and nuts (14).

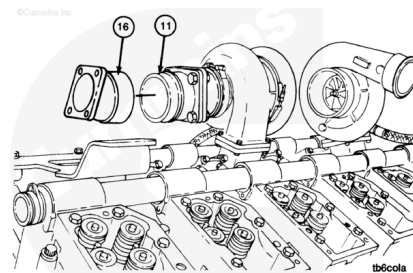


Use anti-seize compound to coat the slip joints of the exhaust pipe (11) and the turbocharger connection (16).

Install the turbocharger connection (16) on the exhaust pipe (11).

After all interstage connections are in place, there **must** be a minimum of 3.18 mm [0.125 inch] expansion space between connections (11 and 16).

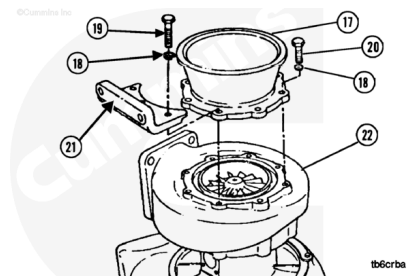
The gap can be adjusted by moving the low-pressure turbocharger or low-pressure turbocharger mounting bracket.



Use anti-seize compound to coat the bore on the low-pressure turbocharger (22) for the exhaust outlet connection (17), and the threads of the capscrews (19 and 20).

The bracket (21) **must** be oriented to provide proper turbocharger alignment.

Note : The two capscrews (19) are longer than the six capscrews (20).



Install the exhaust outlet connection (17), lock washers (18), and capscrews (19 and 20).

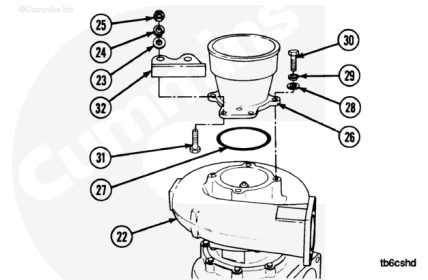
Torque Value:

Exhaust Outlet Capscrews 40 n•m [30 ft-lb]

Install the two capscrews (31), bracket (32), heavy flat washer (23), lock washers (24), and nuts (25) on the air intake connection.

Install the o-ring seal (27) in the groove on the connection (26). Use engine oil or Lubriplate® to hold the seal in the groove.

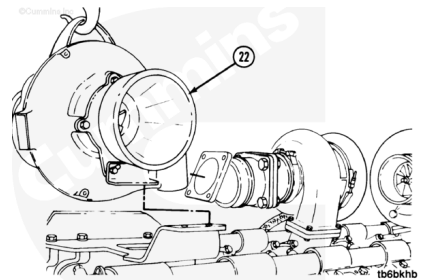
Install the connection (26), four flat washers (28), lock washers (29), and capscrews (30), on the low-pressure turbocharger. Tighten **only** the four capscrews (30).



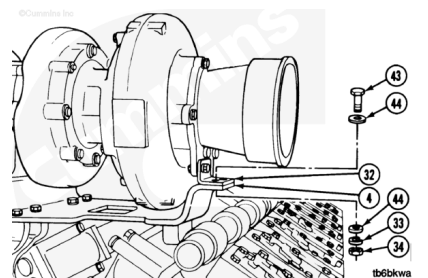
Torque Value:

Air Intake Connection Capscrews 40 n•m [30 ft-lb]

Use a hoist and lifting strap to position the turbocharger and to align the capscrew holes in the brackets.

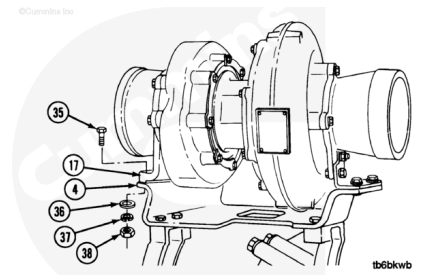


Install the capscrews (43) through the heavy flat washer (44) and both brackets (32 and 4). Install the heavy flat washers (44), lock washers (33) and the nuts (34).



Use anti-seize compound to coat the threads of the capscrews (35).

Install the two capscrews (35) through the exhaust outlet connection (17) and the bracket (4). Install the heavy flat washers (36), lock washers (37), and nuts (38).



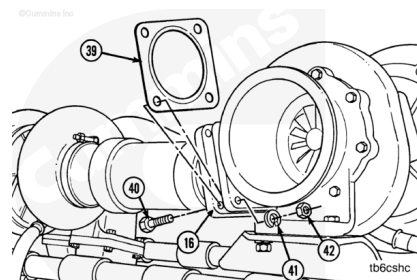
The raised bead on the gasket (39) **must** be toward the turbocharger.

Install the gasket (39) between the turbocharger turbine housing and the exhaust connections (16).

Use anti-seize compound to coat the threads of the capscrews (40).

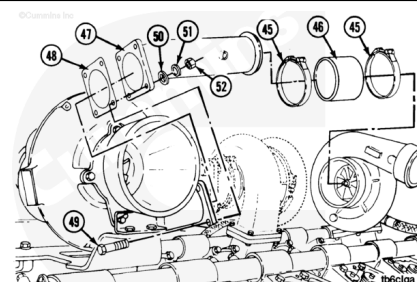
Install the four capscrews (40) through the holes in the turbine housing, gasket, and connection.

Install the lock washers (41) and nuts (42).



Slide one T-bolt clamp (45) and hose (46) over the end of the intake pipe (47). Slide the hose on far enough to install the pipe. Install the second clamp (45) over the compressor outlet.

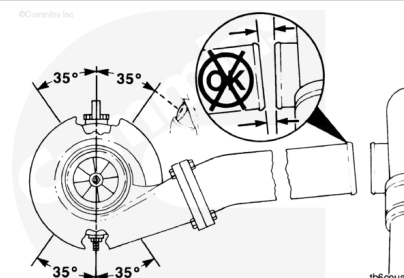
Position the intake pipe (47) between the turbochargers and insert the gasket (48). Secure the flanges with four bolts (49), flat washers (50), lock washers (51), and nuts (52).



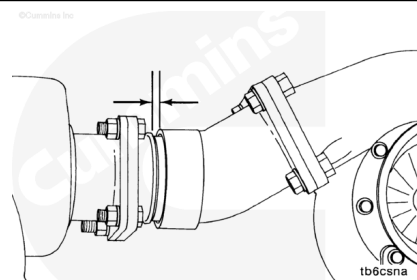
Note : Some of the compressor housing has been removed for clarity.

Check the angle of the oil drain. Check the alignment of the compressor housing to the air inlet pipe.

If the angle or alignment is **not** correct, the bearing housing **must** be rotated in relation to the turbine housing. See Turbocharger, Compressor Housing Alignment, Adjust and Turbocharger, Oil Drain Angle Alignment, Adjust, described in this section.



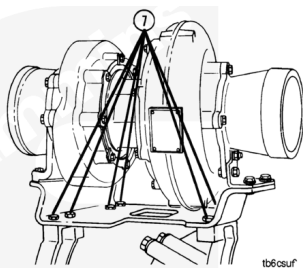
Prior to any of the capscrews being tightened, check to be sure there is a minimum of 3.18 mm [0.125 in] expansion space at the slip joints on both exhaust interstage connections.



Tighten the six capscrews (7) that hold the bracket to the supports.

Torque Value:

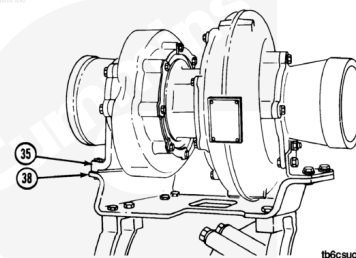
Support Bracket Capscrews 95 n•m [70 ft-lb]



Tighten the two cap screws (35) and nuts (38) that hold the turbocharger exhaust outlet connection to the bracket.

Torque Value:

Bracket Cap screws 95 n•m [70 ft-lb]

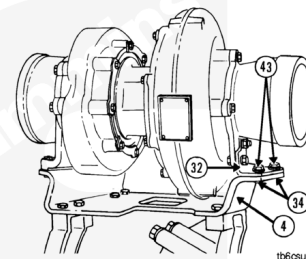


The bracket (32) **must** be flat against the flange on the air inlet connection.

Tighten the two cap screws (43) and nuts (34) that hold the bracket (32) to the support bracket (4).

Torque Value:

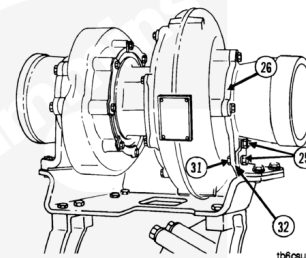
Air Inlet Connection Cap screws 40 n•m [30 ft-lb]



Tighten the two cap screws (31) and nuts (25) that hold the bracket (32) to the connection (26).

Torque Value:

Bracket Connection Cap screws 40 n•m [30 ft-lb]



Tighten the four cap screws (40) and nuts (42) that hold the exhaust connection to the turbine housing.

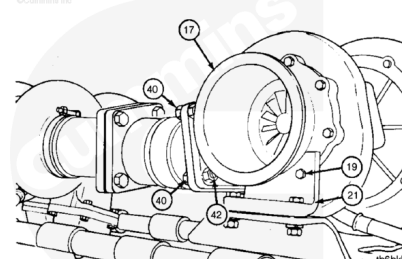
Torque Value:

Exhaust Connection Cap screws 95 n•m [70 ft-lb]

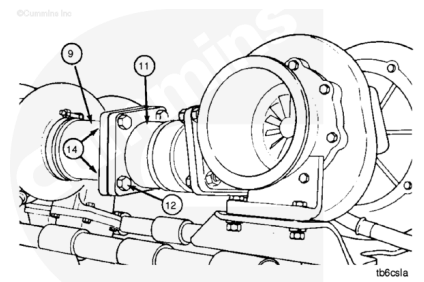
Tighten the three cap screws (19) that hold the bracket (21) to the exhaust outlet connection (17).

Torque Value:

Exhaust Outlet Connection Cap screws 40 n•m [30 ft-lb]



Tighten the four capscrews (12) and nuts (14) that hold the two exhaust connections (9 and 11).



Torque Value:

Exhaust Outlet Connections 95 n•m [70 ft-lb]

Tighten the v-band clamp.

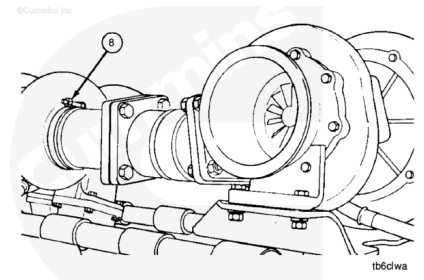
Torque Value:

V-Band 9 n•m [80 in-lb]

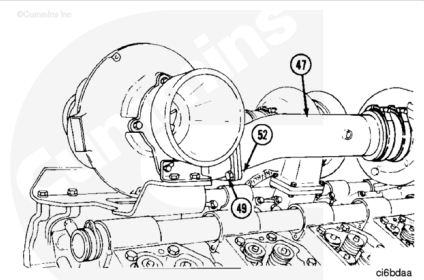
Use a mallet. Tap all around the clamp to make sure it is seated.

Tighten the clamp nut to the same torque value.

Continue tapping and tightening the clamp until the nut will **not** turn at the specified torque after tapping.



Tighten the four capscrews (49) and nuts (52) that hold the air intake pipe (47) to the turbocharger compressor outlet.

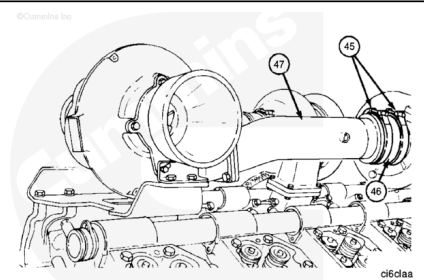


Torque Value:

Intake Pipe Capscrews 40 n•m [30 ft-lb]

⚠ CAUTION ⚠

The hoses and T-bolt clamps of the air intake system are of heavy duty design to withstand temperatures and pressures to which they are exposed. Only specified parts are to be used. The clamps can be removed and tightened again a maximum of three times. The torque retention on the nut will be beyond reuse after this. At this time, a second nut **MUST** be added to double lock the nut or a new clamp **MUST** be used.



Center the hose (46) over the high-pressure turbocharger compressor inlet and the pipe (47). Secure the two T-bolt clamps (45).

Note : Do not position the clamps over the hose beads. The clamps must be positioned between the hose

beads and the end of the hose. A portion of the hose **must** extend outside the clamp.

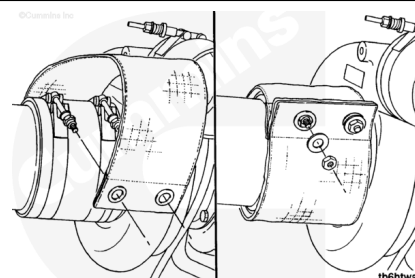
Tighten the two T-bolt clamps (45) on the air intake pipe (47) and the turbocharger compressor inlet.

Torque Value:

Intake Pipe Clamps 8 n•m [71 in-lb]

These clamps need to be tightened again after the engine has operated a few hours.

Wrap the interstage air hose with a heat shield (blanket). The blanket is secured by placing the eyelets that are located on each end of the blanket over the hose clamp T-bolts. Use flat washers and nuts on the T-bolts to retain the heat shield.



Install the gasket (55), plate (56), lock washers (57), and capscrews (58) for the low-pressure turbocharger oil inlet.

Torque Value:

Low-Pressure Turbocharger Oil Inlet 25 n•m [18 ft-lb]

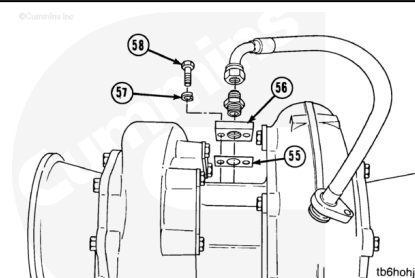
Install the oil inlet fitting.

Torque Value:

Oil Inlet Fitting 35 n•m [26 ft-lb]

Connect the turbocharger oil inlet hose.

The other end of the oil inlet hose **must** be connected to a fitting in the cylinder block. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)



Install the gasket (59), the hose (60), the lock washers (61), and the capscrews (62) for the low-pressure turbocharger oil drain.

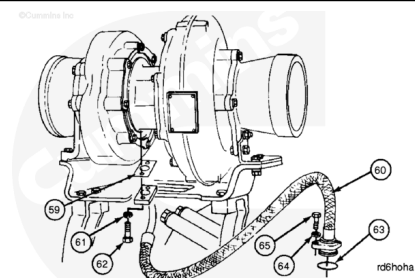
Torque Value:

Turbocharger Oil Drain 25 n•m [18 ft-lb]

Install the seal (63) on the hose flange. Use engine oil to lubricate the seal. Install the flange, lock washer (64) and the capscrews (65) in the proper holes in the cylinder block.

Torque Value:

Hose Flange 40 n•m [30 ft-lb]

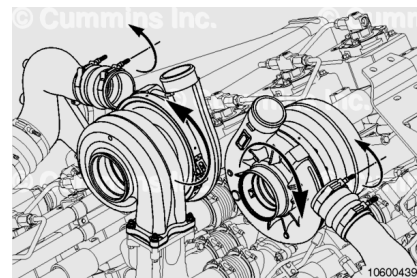


Note : The turbocharger oil drain line **must** run downhill from the turbocharger to the cylinder block.

with Electronically Actuated Injector

⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The word “OUT”, or raised bead on the gasket **must** be positioned toward the turbocharger.

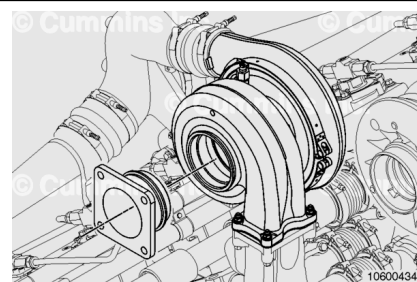
Lift the high-pressure turbochargers, using approved lifting straps, and place them onto the middle mounting pads with the air outlets facing outboard.

Install the capscrews, washers, and nuts, and tighten.

Torque Value:

Turbocharger Mounting Capscrews and Nuts 45 n·m [33 ft-lb]

Install the exhaust outlet and clamp.

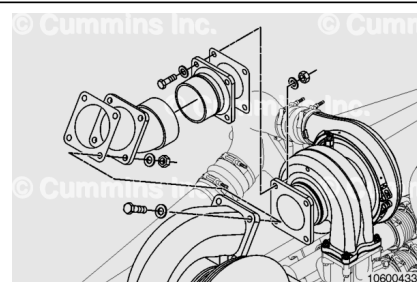


Apply anti-seize compound, Part Number 3824397, or equivalent to both ends of the exhaust interstage piping flanges.

Install both halves of the exhaust interstage using new gaskets.

Apply anti-seize compound, Part Number 3824397, or equivalent to the mounting capscrews.

Tighten the exhaust outlet connection clamp.



Torque Value:

Hose Clamp 8 n·m [71 in-lb]

Install the capscrews, washers and nuts finger tight to allow for alignment.

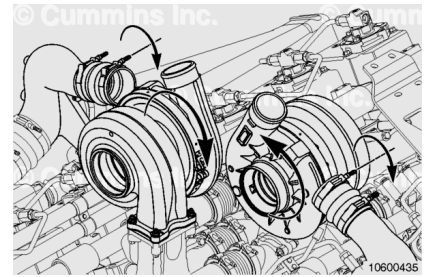
Install the hose and clamps (with tightening screws facing downward) to the inlet side of the turbocharger.

Tighten the hose clamp closest to the turbocharger inlet.

Torque Value:

Hose Clamp 8 n•m [71 in-lb]

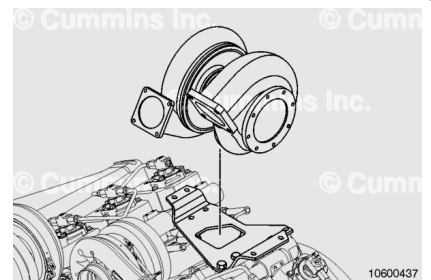
Install the intake air piping into the hose and loosely tighten the clamp to allow for alignment.



⚠ WARNING ⚠

The component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Lift the low-pressure turbochargers, using approved lifting straps, onto the mounting pads and align them with the interstage air transfer connections.



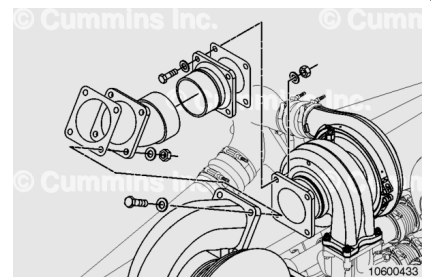
Install new steel gaskets between the outer low-pressure turbocharger and the outer low turbocharger exhaust connection. Check that the raised beads on the gaskets are facing the outer turbocharger.

Install the capscrews, washers and nuts in the same orientation noted prior to removal and hand tighten.

Check the alignment and tighten the capscrews.

Torque Value:

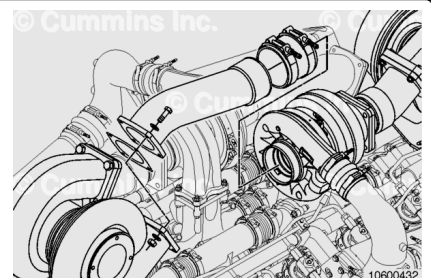
Exhaust Connection Capscrews and Nuts 95 n•m [70 ft-lb]



Install a new paper gasket between the air intake connection and the turbocharger and install the capscrews, washers and nuts. Hand tighten **only** to allow for alignment.

Add the p-clip to the top right capscrew. Do **not** tighten this bolt until the oil supply line is installed.

Once alignment is checked, tighten the capscrews.



Torque Value:

Air Intake Connection Capscrews and Nuts 48 n•m [35 ft-lb]

Check that all air transfer connections are aligned and tighten the interstage hose clamps.

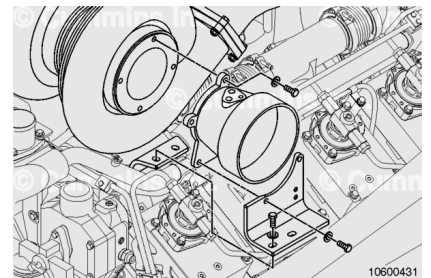
Torque Value:

Interstage Hose Clamp 8 n•m [71 in-lb]

Install the air intake and bracket onto the low-pressure turbocharger and trolley assembly.

Apply anti-seize compound, Part Number 3824394, or equivalent, to the mounting capscrews and install.

Tighten all capscrews with the exception of the two capscrews **not** connected to the mounting bracket.

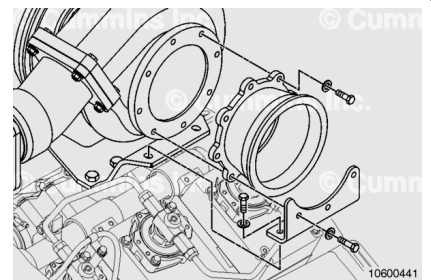


Torque Value:

Air Intake Capscrews 48 n•m [35 ft-lb]

Apply anti-seize compound, Part Number 3824394, or equivalent, to the rear turbocharger outlet flange and install the exhaust adapter and bracket onto the low-pressure turbocharger and trolley assembly.

Apply anti-seize compound, Part Number 3824394, or equivalent, to the eight mounting capscrews and install.



Torque Value:

Exhaust Adapter Capscrews 48 n•m [36 ft-lb]

Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Turn the air ON to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Connect the batteries. Refer to the OEM service manual.

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the oil drain lines. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/28/28-010-045-tr.html>).
- Install the coolant supply and return lines. Refer to Procedure 010-041 in Section 10 (</qs3/pubsys2/xml/en/procedures/28/28-010-041-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>).
- Turn the air ON to the air starter, if equipped. Refer to Procedure 012-022 in Section 12.

(/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)

- Connect the batteries. Refer to the OEM service manual.

Last Modified: 11-Aug-2021
